Healthcare Privatization and Patient Health: Evidence from Puerto Rico *

Dayanara Diaz Vargas†

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Abstract

Public health systems increasingly buy care from private providers. Yet little is known about how patients fare when a publicly financed health system moves entirely from public provision to private management. To study this shift at full scale, I examine Puerto Rico's 1994–2000 health reform, which changed facility ownership, provider payment incentives, and eligibility. Exploiting the reform's staggered rollout across counties, I find that it raised all-cause mortality by 3.7 percent. The increase is accompanied by lower ambulatory-care employment, fewer hospitals, and lower use of early prenatal care, consistent with reduced provider capacity and access to timely care as possible mechanisms. These findings suggest that a broad shift toward privately managed care may harm patient health.

JEL classification: I11, I18, H51, H75, L33.
Keywords: healthcare privatization; managed care; Puerto Rico

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†Northeastern University

1 Introduction

Health spending has risen steadily as a share of GDP across OECD countries, averaging 9.3 percent in 2024, roughly double its 1970 level, and reaching 17.2 percent in the United States (Organisation for Economic Co-operation and Development, 2025). Governments finance most of this spending. Across the OECD, government and compulsory schemes cover close to three-quarters of health costs, and in the United States Medicare, Medicaid, and the Children's Health Insurance Program (CHIP) together accounted for about 7 percent of GDP in 2024 (Medicaid and CHIP Payment and Access Commission, 2026; Centers for Medicare and Medicaid Services, 2025a). To reduce costs, governments have increasingly relied on the private sector to manage their healthcare. There's a rich literature that examines these effects at small scale and for specific aspects of privatization (Fabre and Straub, 2023). I contribute to this literature by studying this shift at an unusually large scale, examining Puerto Rico's 1994 health reform, which transferred the management of its extensive public healthcare system to the private sector.

Before the reform, the Commonwealth ran a system of direct public provision, in which it (i) owned the facilities, (ii) employed salaried providers, and (iii) provided universal coverage, since any resident could receive care in its facilities. The reform changed all three. It (i) sold or leased the facilities to private providers, (ii) replaced government provider salaries with private-sector capitated contracts, and (iii) replaced universal coverage with a government health plan for residents below 200 percent of the Puerto Rico poverty line, administered by private Medicaid managed-care organizations (MCOs) (Commonwealth of Puerto Rico, 1993). In doing so, the government sought to reduce costs by shifting publicly financed care from public to private provision. The reform reached the island's 78 counties in six cohorts between 1994 and 2000.

I build a county-year panel for Puerto Rico's 78 counties from four sources. The Puerto Rico Planning Board provides annual population and deaths by cause (Junta de Planificación

de Puerto Rico, 2026). The Department of Health's annual vital-statistics reports, digitized for this paper for 1990 to 2002, add births by hospital ownership, prenatal care, and birth weight (Puerto Rico Department of Health, 2025). The Quarterly Census of Employment and Wages measures ambulatory health care employment, establishments, and wages (U.S. Bureau of Labor Statistics, 2025), and the Provider of Services files of the Centers for Medicare and Medicaid Services identify Medicare-certified hospitals (Centers for Medicare and Medicaid Services, 2025b). Mortality for mainland counties in states that had not adopted Medicaid managed care by 2000 provides a second comparison group (National Center for Health Statistics, 2026).

I compare counties after the reform reaches them with counties it has not yet reached, using the estimator of Callaway and Sant'Anna (2021) to account for heterogeneous treatment effects across cohorts and years. Because Puerto Rico is an island, no contiguous USA states provider is within reach for routine care, so cross-state-border patient flows are absent. Puerto Rico's federal Medicaid funding is also subject to an annual cap, leaving the Commonwealth responsible for additional spending once the federal allotment is exhausted (Mitchell, 2025). This financing structure offers two advantages for our analysis. First, when the cap is binding and the allotment remains unchanged, increases in federal matching rates or expansions in Medicaid eligibility do not generate additional federal funds, limiting one channel through which federal policy changes could confound the estimated effects of the reform. However, since patients could still move between counties on the island, and Section 7 examines commuting patterns across reform cohorts, showing that approximately 70% of residents' commutes to work remain within their own cohort (Appendix Figure A4, Panel (a)). To the extent that travel for healthcare follows similar patterns, this finding suggests that most patients also seek care within their own reform cohort.

The main outcome is all-cause mortality. The reform raised it by 26.96 deaths per 100,000 residents (s.e. 10.32), or 3.7 percent of the pre-treatment mean of 724.2 (Table 3). Ambulatory health care employment grew across the island over the decade (Table 1), but in counties

that entered it ran 0.2663 jobs per 1,000 residents below its counterfactual path (s.e. 0.1494), 9.6 percent of the pre-treatment mean and significant at the 10 percent level. Establishments and certified hospitals per resident are also lower relative to the counterfactual (Table 4). These estimates point to reduced provider capacity as one mechanism, and their direction matches the staffing reductions at privatized U.S. hospitals (Duggan et al., 2023).

Because Puerto Rico changed all three components across much of its public system, this paper provides the first causal evidence on moving a public health system's delivery to private management.[1] These results are informative for governments that increasingly continue to shift care from public to private providers. Government-controlled hospital capacity in the United States fell by 42 percent between 1983 and 2019 (Duggan et al., 2023), and the Veterans Health Administration, which owns and staffs more than 1,700 sites, raised its spending on outsourced care from $7.9 billion in 2014 to $17.6 billion in 2021, in 2021 dollars, or from about 12 to 20 percent of its medical-care spending (Panangala et al., 2025; Congressional Budget Office, 2021). In England, the health department bought £24.1 billion of care outside the National Health Service in 2024–25 (National Audit Office, 2026), and in New Zealand the private share of publicly funded elective procedures rose from 8.6 percent in 2011–12 to 16.7 percent in 2023–24 (Radio New Zealand, 2026).

Prior studies vary one of these components at a time. Some compare public and private providers within a system that keeps both (Chan et al., 2023; Goodair and Reeves, 2022) or follow the privatization of a set of hospitals (Duggan et al., 2023; Tiemann and Schreyögg, 2012), which varies who runs the facilities; others change how providers are paid or who is covered while providers stay private (Duggan, 2004; Duggan et al., 2018; Finkelstein et al., 2012; Aizer et al., 2007). In each setting the rest of the health system stays in place, so the estimates can be affected by spillovers onto the part that did not change, and none of them estimates the effect of changing a public delivery system as a whole. Coube et al. (2026)

[1] Georgia's transfer of public healthcare to private management, beginning in 2007, is studied qualitatively by Tvaliashvili et al. (2025).

test for such spillovers when Brazilian governments contract out the management of public hospitals and find none onto neighboring hospitals, at the scale of adoption they observe. Puerto Rico instead changed all three components across its public delivery system, county by county, and this paper estimates the effect of that bundle.

Evidence on privatization and patient outcomes is mixed. For-profit conversions of U.S. hospitals cut staffing and raised patient mortality one to two years later (Picone et al., 2002), and private ambulance providers in Stockholm improved on contracted dimensions while raising three-year mortality (Knutsson and Tyrefors, 2022). In England, more outsourcing of publicly funded care to for-profit providers after 2012 is associated with higher treatable mortality, although the authors state that their design cannot identify a causal effect (Goodair and Reeves, 2022). When U.S. government hospitals passed to private control, 30-day mortality among their elderly patients rose, as did near-elderly mortality in the markets where Medicaid admissions fell most (Duggan et al., 2023), while German hospitals converted to for-profit status became more efficient with no measured loss of quality (Tiemann and Schreyögg, 2012). Dually eligible veterans fare comparably or better in Veterans Affairs care than in private care (Chan et al., 2023), and a meta-analysis of 13 longitudinal studies finds adverse or mixed effects of conversion to private ownership on quality of care (Goodair and Reeves, 2024). On the payer side, Medicaid managed care reduced prenatal care and raised low birth weight and neonatal death among the mothers subject to it (Aizer et al., 2007), while Medicare Advantage plans restricted hospital use with no measurable change in mortality (Duggan et al., 2018). The outcome in this paper is mortality among all residents of the counties the reform reached.

The estimate stays positive and significant when counties that differ in six baseline characteristics may follow different mortality trends, whether the characteristics enter one at a time or all together (Table 6), when counties are weighted by population (Table 7), and under three other estimators that allow treatment effects to vary across cohorts and years (Table 8). In logs it is smaller and significant only at the 10 percent level. A comparison with

mainland counties in states that had not adopted Medicaid managed care by 2000 shows a coefficient that is consistent with our main findings but lacks statistical significance at the traditional levels of confidence[2](see Section 7).

Section 2 describes the setting, Section 3 the theoretical framework, Section 4 the data, and Section 5 the identification strategy. Section 6 presents results on who delivered care, health outcomes, healthcare supply, and mortality by cause, Section 7 assesses robustness, and Section 8 concludes.

2 Setting

2.1 Before the reform

From the mid-1950s, Puerto Rico managed its public healthcare program through the integrated, regionalized system described by Arbona and Ramírez de Arellano (1978). Government facilities and salaried staff provided local care through 78 Centers of Diagnostic and Treatment (CDTs), one per county, which offered primary and emergency care, and subregional and regional hospitals supplied inpatient and specialized outpatient care (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). The system was designed for the medically indigent (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005), but any resident could receive care in its facilities, so coverage was universal. In US terms, it resembled the Veterans Health Administration, which also owns and staffs its facilities (Panangala et al., 2025), except that Puerto Rico's system was open to every resident. Abroad, Britain's National Health Service, whose hospitals are publicly owned, follows the same model of public delivery (Bloom et al., 2015). Appendix E describes the services each level provided and how patients were referred

[2]This estimate is close to the baseline but imprecise, due to its lack of variance. The estimator reweighs mainland counties toward Puerto Rico's 1990 poverty rates, and few mainland counties are comparable: island county poverty ranges from 34 to 79 percent, while 90 percent of mainland counties fall between 6.6 and 31.4 percent. The estimate therefore rests on a small comparison group (see Table 7, column 6).

between levels.

Puerto Rico was exempt from the federal requirement that Medicaid recipients be free to choose their providers, so its Medicaid paid only for care in Commonwealth and municipal facilities and made no payments to private providers (Pagán-Berlucchi and Muse, 1983; Appendix E). This held in each county until the reform reached it. Public healthcare coexisted with an extensive private sector, in which access to private hospitals depended on insurance or the ability to pay (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). In 1990, the public system operated roughly 40 percent of hospital beds and served about 1.8 million residents, or 51.4% of the population (Instituto de Administración y Política de Salud, 1996; Appendix E). During that time, Puerto Rico had 13 public and 56 private hospitals (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

In 1979, the Department of Health employed 57.3 percent of the island's practicing health professionals (Pagán-Berlucchi and Muse, 1983). In 1992, the Planning Board counted 8,443 licensed physicians and 16,848 registered nurses on the island (Junta de Planificación de Puerto Rico, 2026; Appendix Table A11). In 1993, private ambulatory health care providers employed 15,822 workers (Appendix Table A2), and the county mean of annual pay per worker was $21,283 in 2025 dollars (Table 1).

Federal Medicaid funds, available in Puerto Rico beginning in 1965 (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005), financed part of public-system expenditures, and the Commonwealth covered the remainder.[3] In fiscal year 1979, federal Medicaid payments, capped since 1968, covered 17.3 percent of public health system spending (Pagán-Berlucchi and Muse, 1983). Act 72 of 1993 attributed fiscal pressures to insufficient budgets, rising technology and supply costs, bureaucratic growth and centralization, and political interference in the Department of Health, which served residents

[3]Federal Medicaid funding for Puerto Rico is subject to a capped territorial allotment (Medicaid and CHIP Payment and Access Commission, 2020).

unable to afford private care (Commonwealth of Puerto Rico, 1993). The reform sought to reduce costs for the government through contracts with insurers and providers while preserving medical and hospital access for low-income patients (Commonwealth of Puerto Rico, 1993).

2.2 Reform implementation

The reform replaced this integrated public delivery system with managed care delivered mainly by private providers (Pan American Health Organization, 2007). In each county it reached, (i) the county's CDT and the area and regional hospitals serving it passed to private providers through sales and long-term leases, (ii) contracts with private managed-care plans replaced government salaries as the way providers were paid, and (iii) universal access at the public facilities gave way to enrollment in the government health plan for residents below 200 percent of the Puerto Rico poverty line.

Act 72 of September 7, 1993, reorganized public healthcare around Medicaid managed care (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). It created the Puerto Rico Health Insurance Administration to administer coverage funded through Medicaid and contract with health maintenance organizations (HMOs) (Commonwealth of Puerto Rico, 1993).[4] Private MCOs managed provider networks and paid for covered services, and providers delivered care. Eligible residents enrolled in a plan and became its cardholders, entitled to covered care from the providers in its network (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Public financing and government decisions over eligibility and benefits remained under this arrangement.

The government paid insurers capitated premiums, which consisted of predetermined amounts per enrollee for specified services over a fixed period. Insurers then negotiated

[4]Appendix B reports the reformed system's funding by source, including federal Medicaid.

payment terms with network providers (Commonwealth of Puerto Rico, 1993; World Health Organization and Pan American Health Organization, 1998; Appendix E). The goal of this system is to decrease costs, since an insurer benefits when covered care costs less than the premiums it receives. Another way in which insurers decreased costs was by setting primary-care providers as gatekeepers to specialized care, since specialist care, diagnostic tests, and procedures required their referrals (Centers for Medicare and Medicaid Services, 2000, 2001, 2002, 2003). In Figure 1, panel (d), government spending on public delivery falls from 1994 while the premiums paid to MCOs rise, and the sum of the two rises until 1998 and then falls; Appendix B defines the series. For patients obtaining care through the government health plan, access thus depended on self-enrollment, the insurer's provider network, and its referral arrangements.

The government health plan worked like the mandatory Medicaid managed care that California phased in county by county during the 1990s (Duggan, 2004; Aizer et al., 2007). California's recipients had used providers paid fee for service before their county's mandate (Duggan, 2004). In Puerto Rico, the reform also let enrollees use their public coverage with private providers, which Medicaid had not paid before, and the public facilities themselves passed to private providers over the rollout.

Alongside insurance contracting, the government transferred facilities to private providers. The 78 CDTs and eleven area and regional hospitals were sold or transferred under long-term leases, while the Commonwealth retained its two largest public hospitals, the Puerto Rico Medical Center and the Cardiovascular Center (Instituto de Administración y Política de Salud, 1996; Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005; Appendix E).[5] Hospital births shifted toward private providers. Across counties, the mean share of births in government hospitals fell from 65.8 percent in 1990 to 30.6 percent in 2000 (Table 1), which is indicative of the policy's privatization efforts.

[5]Fabre and Straub (2023) include long-term leases among partnership contracts and classify a sale, which transfers title, as privatization. The facility transfers therefore fall on both sides of their definition.

The government health plan offered broad benefits and low cost sharing to residents with incomes below 200 percent of the Puerto Rico poverty line (Centers for Medicare and Medicaid Services, 2000), and enrollment eventually reached close to half the population (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). The benefit package covered hospital, ambulatory, pharmacy, dental and mental-health care without pre-existing-condition exclusions (Centers for Medicare and Medicaid Services, 2000), and Appendix B lists the covered services. Income-dependent cost sharing ranged from \$1 to \$4 per visit and \$0.50 to \$3 per prescription (Instituto de Administración y Política de Salud, 1996; World Health Organization and Pan American Health Organization, 1998).

The government selected MCOs through competitive bidding, with one MCO serving each health region, a group of counties (World Health Organization and Pan American Health Organization, 1998; Brugueras Fabre, 2002). The reform entered region by region, the first counties in February 1994 and San Juan in 2000 (Brugueras Fabre, 2002; Figure 1, panel (a); Appendix E). The six cohorts comprise 18 counties in 1994, 27 in 1995, 7 in 1996, 13 in 1997, 12 in 1998, and San Juan alone in 2000. A county's cohort year is the year the reform reached it, measured as the first year its residents were enrolled in the government health plan. The cohort year does not date the transfer of a county's facilities or the change in its providers' payment, so the estimates measure the effect of the reform's arrival together with those changes as they reached each county. Enrollment rose from 130,392 residents in 1994 to 1,795,067, or 47% of the population, when the rollout ended in 2000, out of 1,866,965 residents eligible that December (Administración de Seguros de Salud de Puerto Rico, 2026).

By 2003, 7 percent of residents lacked a health plan, and the commission that evaluated the health system in 2005 attributes the growth of the uninsured population to government facilities that shrank or disappeared (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Appendix E reports take-up and the composition of coverage.

The reform worked mainly on the supply side of the market for care, through who ran the facilities and how providers were paid. About half of residents were in a public program or had no coverage, 51 percent in 1990 and 47 percent in 2003, and the rest held private insurance, Medicare or the public employees' plan (Appendix E). Enrollment at the end of the rollout was close to the 1.8 million residents the public system served in 1990, so the number of residents whose care the government financed changed little over the rollout, apart from incomplete take-up in its first years and the coverage gap above. Cost sharing was low, so the reform did not rely on the prices patients paid to limit their use of care. Each plan received a fixed premium per enrollee and made the primary-care physicians who controlled referrals bear financial risk for the care other providers gave their enrollees (Appendix E). In the terms of Ellis and McGuire (1993), the reform relied on supply-side cost sharing. Copayments and premiums were set administratively, so prices did not adjust to clear the market, and plans rationed any excess demand through referrals. Part of the shift of births to private hospitals reflects the hospitals themselves passing to private providers (Section 6.1). The estimates measure the reform as a whole (Section 5).

In the health labor market, the government, which in 1979 had employed most of the island's health professionals, reduced budgeted positions in its health delivery and regulatory agencies from 26,905 in fiscal year 1996 to 18,062 in 1999 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998). By 2000, private ambulatory employment had risen to 24,815 jobs and the county mean of annual pay per worker to $23,742 (Appendix Table A2; Table 1). The employment count includes any CDT staff the new providers kept on (Section 4.2). The Planning Board counted 7,144 licensed physicians and 12,145 registered nurses in 1998 (Appendix Table A11). Higher private employment and pay are consistent with private demand for health labor shifting outward as government positions were cut, and the lower licensed counts with a labor supply that did not grow to meet it. These island series also reflect everything else that changed over the decade, and Section 6.3 estimates the reform's effect against counties that had not yet entered.

3 Theoretical framework

I adapt Hart et al. (1997) to a bundle of healthcare services. When contracts cannot specify every production decision or dimension of care, residual control rights determine who can implement changes the contract leaves open (Grossman and Hart, 1986; Hart et al., 1997). The reform moved these rights, and the savings from cutting costs, from the government to private managers. In the model the move raises cost-cutting on dimensions of care that no contract covers, which can harm patients, and raises quality improvement only where public managers kept little of an improvement's value. I represent the MCO and its providers as one delivery manager, to whom control can pass through a sale, a lease, or an operating agreement; this assumes that the MCO passes the incentive of its capitated payment through to its providers. The model compares public and private control while holding eligibility, cost sharing, and covered benefits fixed; Section 2.2 describes what stayed public.

For each service s, the manager chooses two efforts that no contract can specify, $e_s \geq 0$ and $i_s \geq 0$. Cost-reducing effort e_s produces a cost innovation that saves production costs $c_s(e_s)$, with $c_s' > 0$ and $c_s'' < 0$, and harms patients by $b_s(e_s)$, with $b_s' > 0$ and $b_s'' \geq 0$, for example by reducing access or treatment intensity on labor-intensive dimensions of care that are hard to verify. In the cross-sector comparison of Fabre and Straub (2023), health and education are the two sectors whose service quality is hard to verify. Quality-improving effort i_s produces a quality innovation that raises the value of care to patients by $v_s(i_s)$, with $v_s' > 0$, at a direct production cost $k_s(i_s)$, so its net benefit $\beta_s(i_s) = v_s(i_s) - k_s(i_s)$ satisfies $\beta_s' > 0$ and $\beta_s'' < 0$. These quantities and the manager's cost of effort, $e_s + i_s$, are measured in the same units of value.

A private manager paid a fixed amount per enrollee keeps any cost savings and, holding control, can implement a cost innovation without the government's approval.[6] The manager

[6]Coube et al. (2026) study Brazil's Organizações Sociais de Saúde (OSS) contracts, which give nonprofit managers control over staffing and procurement in public hospitals but require them to reinvest any budgetary savings. They read the arrangement as public ownership with private management in the sense of Hart et al. (1997). In this model's terms the contract removes the manager's private return to cost containment, and

does not bear the patient harm, which no contract term prices, and nothing in the model lets patients leave a manager whose care deteriorates. Under a fixed payment the manager bears the cost of a quality innovation and is paid for it only by renegotiating the contract with the government, and symmetric bargaining gives the manager half of the innovation's net benefit. Under public control the government holds the residual rights, so a salaried public manager needs its approval for either innovation, and the government values a cost innovation at its savings net of the patient harm. Bargaining then gives the public manager $\lambda/2$ of each innovation's net value, where $\lambda \in (0, 1]$ is the share of that value requiring the incumbent manager's participation; the government can obtain the rest by replacing the manager. The two managers therefore solve

$$\text{Private control:} \quad \max_{\{e_s, i_s\}} \sum_s \left[c_s(e_s) + \tfrac{1}{2}\beta_s(i_s) - e_s - i_s \right], \tag{1}$$

$$\text{Public control:} \quad \max_{\{e_s, i_s\}} \sum_s \left[\tfrac{\lambda}{2}\left(c_s(e_s) - b_s(e_s) \right) + \tfrac{\lambda}{2}\beta_s(i_s) - e_s - i_s \right]. \tag{2}$$

The easier the manager is to replace, the smaller λ and the less of either innovation the public manager captures. Appendix C derives the first-order conditions (C.3) and the ordering of efforts they imply (C.4).

Equation (C.2) defines the first best, which maximizes the savings net of harm plus the net benefit of quality, less effort costs. Relative to it, a public manager does too little of both kinds of effort. A private manager pushes cost-reducing effort past the first best and improves quality more than a public manager whenever $\lambda < 1$, though less than the first best. Equation (C.5) writes mortality as falling with the value of care patients receive and rising with the harm from cost-cutting, each weighted by how much the service matters for survival, with other determinants of mortality unaffected by who controls care.

At $\lambda = 1$ a public manager keeps the same half of a quality innovation as a private one, so both make the same quality effort, and privatization changes only the cost-cutting, which

they find higher admissions with no detectable change in risk-adjusted inpatient mortality.

raises mortality. When public managers were easier to replace ($\lambda < 1$), privatization also brings a quality gain that can offset the harm, and the sign of the mortality change depends on which is larger. Proposition 1 in Appendix C states this comparison formally and gives the following hypothesis.

Hypothesis 1 *Moving residual operational control of care from public to private managers raises mortality if, across services weighted by their importance for survival, the harm to patients from the additional cost-cutting exceeds the value to patients of the additional quality improvement. The condition holds when public managers were hard to replace ($\lambda = 1$) and for λ close enough to* 1.

A fall in mortality would indicate that the quality gain dominated. Because the model holds eligibility fixed, the hypothesis concerns the control and payment components of the reform, and the estimates measure the whole bundle (Section 2.2).

The shift of births from government to private hospitals (Section 6.1) shows that control of care moved, and all-cause mortality (Section 6.2) is the outcome the hypothesis signs. Early prenatal care measures access to timely care, which cost-cutting can reduce and the change in eligibility could also move. Employment, establishments, and certified hospitals (Section 6.3) are inputs, which cost-reducing effort saves and quality improvement can either require or save, as in equation (C.7); the model therefore does not sign their change, and a shortfall measures neither effort nor efficiency. The model is static, and equation (C.8) shows how a lasting change in care lowers health progressively over time. Because λ and the survival weights are not observed, the condition in the hypothesis cannot be tested, and the estimates indicate which case of Proposition 1 the evidence fits.

4 Data

4.1 Health outcomes

I combine administrative and vital-statistics data for Puerto Rico's 78 counties. The Puerto Rico Planning Board provides annual population, births, all-cause and cause-specific deaths, and county characteristics (Junta de Planificación de Puerto Rico, 2026)[7]. The all-cause mortality panel has all 78 counties in every year from 1990 to 1999, and the tables note where a cause or a birth outcome is missing in a year. Table 1 summarizes the main outcomes, and Appendix Table A1 reports baseline characteristics by cohort.

All-cause and cause-specific mortality are deaths per 100,000 residents, without age adjustment, using annual intercensal population from the Planning Board in Puerto Rico and SEER on the mainland. Infant mortality is infant deaths per 1,000 live births, and maternal mortality is maternal deaths per 100,000 live births. The birth outcomes are shares of resident live births, as the vital-statistics reports define them: early prenatal care is care begun in the first trimester of pregnancy, and low birth weight is a weight below 2,500 grams regardless of gestational age, a category that includes very low birth weight, below 1,500 grams (Puerto Rico Department of Health, 2025). The government-hospital birth share is births in a government-operated hospital over all births. Monetary outcomes use constant 2025 dollars and the all-items consumer price index (U.S. Bureau of Labor Statistics, 2026). In 1993, the year before the reform, 99.8 percent of births and 54.5 percent of deaths on the island occurred in a hospital (Puerto Rico Department of Health, 2025). Because nearly every birth took place in a hospital, the ownership of the hospital where a birth occurred tracks the shift in supply from public to private providers. Patients rarely needed to leave

[7]Digitized annual vital-statistics reports for 1990–2002 add births by hospital ownership, prenatal care, birth characteristics, and early observations for mortality causes incompletely recorded by the Planning Board (Puerto Rico Department of Health, 2025). I retain source flags and leave unobserved counts missing. Overlapping series have correlations of 0.99–1.00. Checks against printed subtotals and island totals assess transcription consistency; comparisons between publications assess measurement agreement, not necessarily independent registration. The data-construction appendix documents these checks, exceptions, and classification changes.

their cohort for care. Every cohort had at least one hospital, and in 2000, 70.7 percent of resident workers were employed in a county of their own cohort (Appendix Figure A4). That year, 85.6 percent of deaths occurred in a hospital or at home (Puerto Rico Department of Health, 2025), so county mortality reflects access to hospital and preventive care where residents live.

4.2 Healthcare supply

The Quarterly Census of Employment and Wages (QCEW) records annual employment, establishments, and payroll across the United States and Puerto Rico (U.S. Bureau of Labor Statistics, 2025). It counts jobs covered by unemployment insurance at their workplace, excluding self-employment. I measure healthcare supply in Ambulatory Health Care Services (NAICS 621): physicians' offices, outpatient clinics, and diagnostic services. QCEW publishes no government ambulatory employment for Puerto Rico, so staff of transferred CDTs whom private providers kept on enter observed employment when their facility is transferred, and the employment estimate is net of their entry. I use NAICS 621 rather than the whole health care and social assistance sector (NAICS 62) for three reasons: ambulatory care is the care the capitated contracts paid for and the CDTs provided; NAICS 62 adds social assistance (NAICS 624), which is not healthcare; and NAICS 621 is published as one series, while a total of NAICS 621 to 623 sums parts that are not published in every county and year. The wider sector appears in County Business Patterns, whose health services series includes hospitals (Appendix Table A2, Panel D, and Appendix Figure A1). The outcomes are employment and establishments per 1,000 residents, employees per establishment, and annual payroll per worker in constant 2025 dollars. The QCEW panel has the 59 counties published in every year from 1990 to 2000. Their employment is 97.9 percent of the island's ambulatory employment in 1993 and 96.8 percent in 2000, measured as the panel counties'

employment over the island total for the same year (Table 1).[8] The Centers for Medicare and Medicaid Services Provider of Services files identify hospitals certified for Medicare, the federal health insurance for older people and certain people with disabilities (Centers for Medicare and Medicaid Services, 2025b). The 1991–2005 panel covers all 78 counties, including 47 with no certified hospitals during this period, and the outcome is certified hospitals per 1,000 residents.

4.3 Mainland comparison data

To extend comparisons beyond Puerto Rico's completed rollout, I assemble mainland county mortality from CDC WONDER (National Center for Health Statistics, 2026), population and age composition from Surveillance, Epidemiology, and End Results (SEER) (National Cancer Institute, 2026), and poverty and land area from the 1990 Census (U.S. Census Bureau, 1990). The primary sample excludes states adopting Medicaid managed care by 2000. These counties are the comparison group of the mainland specification in Section 7.3 (Table 7, column 6).

5 Identification strategy

I estimate the effects of the reform using the staggered difference-in-differences framework of Callaway and Sant'Anna (2021). For each adoption cohort, I compare changes in outcomes with those in counties not yet treated. Let Y_{it} denote county i's outcome in year t, G_i its adoption year, and D_{it} an indicator equal to one once county i has been treated by year t. The potential outcomes $Y_{it}(g)$ and $Y_{it}(0)$ correspond to first adopting in year g and to

[8]Two sources assess QCEW coverage. County Business Patterns closely matches pre-reform private ambulatory employment in the available county cross-section (U.S. Census Bureau, 2000). Census microdata count more healthcare workers, with a narrowing gap over the decade (Ruggles et al., 2025). That comparison is island-wide and measures resident workers rather than workplace jobs. The data-construction appendix details differences in industry definitions, reporting, and reference periods. See Appendix Table A2 for more detail.

remaining untreated throughout. The group-time average treatment effect is $\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g]$. Write $\bar{g}$ for the last adoption year and $\mathcal{G} = \mathrm{supp}(G) \setminus \{\bar{g}\}$. Because no county remains untreated throughout, $\mathcal{G}$ excludes the last-treated cohort, which has no valid untreated comparison group.

Without covariates, equation (2.9) of Callaway and Sant'Anna (2021) identifies the group-time effect as

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{i,g-1} \mid D_{it} = 0]\,, \qquad g \in \mathcal{G}, \quad g \le t < \bar{g}, \tag{3}$$

where the second expectation is taken over counties not yet treated by year t, that is, those with $G_i > t$. Two assumptions deliver equation (3). Absent the reform, mortality in counties that entered in year g would have changed between $g-1$ and t by as much as mortality in counties that had not yet entered by t, and the same holds for every other outcome. Entry also changed nothing before it happened, so no outcome moved in anticipation. A county's cohort year is the year the reform reached it (Section 2.2), so $\mathrm{ATT}(g,t)$ is the effect of the reform's arrival together with the facility and payment changes as they reached the county by year t.

The last county entered in 2000, so $\bar{g} = 2000$. By 1998 the other 77 counties had entered, leaving San Juan as the only comparison county in 1998 and 1999. I therefore identify effects for the earlier cohorts through 1999 and write $\bar{t} = \bar{g} - 1 = 1999$ for the last identified year. San Juan is only a comparison county, since its own post-adoption effect is not identified. The reform entered region by region on the schedule documented by Brugueras Fabre (2002), and the cohorts differ at baseline, most in population and population density (Appendix Table A1).

For each specification I also report an event study. Let $e = t - g$ denote event time, the number of years elapsed since a county adopted the reform, and recall that G is the year in which a county is first treated. Summarizing how effects vary with length of exposure

therefore amounts to averaging the $\mathrm{ATT}(g,t)$ across cohorts at a common value of e, which is the aggregation in equation (3.4) of Callaway and Sant'Anna (2021),

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq \bar{t}\} \, \mathrm{P}(G = g \mid G + e \leq \bar{t}) \, \mathrm{ATT}(g, g + e). \tag{4}$$

This is the average effect e years after adoption across the cohorts ever observed at that time period, and $e = 0$ gives the effect of the policy in the adoption year itself. The indicator restricts the sum to identified group-time effects (Callaway and Sant'Anna, 2021). The weights are cohort shares among the cohorts observed at event time e, so the set of cohorts behind $\theta_{es}(e)$ changes with e: the longest horizons rest on the 1994 and 1995 waves, and the earliest leads on the latest cohorts.

The leads, the estimates for $e < 0$, compare counties before they entered with counties that had not yet entered. Leads near zero are consistent with the first assumption but cannot establish it, because the assumption concerns the years after entry. Some leads exclude zero in the plotted 95 percent simultaneous bands. The all-cause lead at $e = -8$ does so in Figure 2 and in every panel of Figure 6; the employment leads at $e = -8$ and $e = -7$ and the certified-hospital lead at $e = -6$ do so in Figure 4; and the early-prenatal-care lead at $e = -2$ does so in Figure 3.

The baseline includes no covariates. Because the cohorts differ at baseline, additional specifications in Section 7 use the doubly robust estimand (2.7) of Callaway and Sant'Anna (2021), which allows untreated trends to depend on pre-reform county characteristics. The reported overall coefficient is θ^{O}_{W}, equation (3.10) of Callaway and Sant'Anna (2021), which uses a weighted average of all identified group-time effects with weights proportional to cohort size, so counties contributing more post-reform years receive greater total weight. Inference follows with county-specific bootstrap multipliers and 1,000 replications.

6 Results

First, I consider who delivered care, showing that births moved from government to private hospitals as the reform reached each county. I then report the effects on health outcomes, on healthcare supply, and on mortality by cause.

6.1 Who delivered care

Relative to counties not yet treated, the government-hospital share of residents' live births fell by 17.94 percentage points (s.e. 1.25), 28.0 percent of its pre-treatment mean, and the private-hospital share rose by 17.29 points (s.e. 1.19), 47.5 percent of its mean (Table 2; Figure 1, panels (b) and (c)). Births moved from government to private hospitals as the transfers described in Section 2.2 reached each county.

6.2 Health outcomes

I estimate an increase in all-cause mortality of 26.96 annual deaths per 100,000 residents (s.e. 10.32), equivalent to 3.7 percent of the pre-treatment mean of 724.2 (Table 3).[9] In the event study (Figure 2), the effect grows with exposure, and the leads are near zero except at $e = -8$ (Section 5). The direction matches the higher 30-day mortality of elderly patients after U.S. public hospitals were privatized (Duggan et al., 2023) and the higher three-year mortality under private ambulance providers in Stockholm (Knutsson and Tyrefors, 2022).

The all-cause rate counts residents' deaths. In 1993, the year before the reform, 54.5 percent of deaths occurred in a hospital, 30.5 percent in a government hospital and 24.0 percent in a private one, and 63.6 percent of the deceased had received medical assistance (Puerto Rico Department of Health, 2025). By 2000, as government hospitals passed to private management, 33.5 percent of deaths occurred in private hospitals, 22.1 percent in

[9] Appendix Table A3 reports island deaths by cause and year.

government hospitals and 30 percent at home (Puerto Rico Department of Health, 2025). Homicides and accidents were 3.3 and 4.2 percent of deaths in 1993 (Appendix Table A3), so most deaths outside hospitals were neither. The reform could therefore raise the death rate through the care patients received in hospitals that changed hands and through the ambulatory and preventive care that keeps conditions from worsening before a patient reaches a hospital, which decreased in treated counties (Section 6.3). The place of death does not divide the increase between these two paths, since a patient whose condition went untreated in ambulatory care can still die in a hospital.

Figure 3 reports the event studies for prenatal care and the birth outcomes; the early-prenatal-care lead at $e = -2$ excludes zero (Section 5). The share of births with early prenatal care falls by 2.33 percentage points (s.e. 0.62), or 3.0 percent of its pre-reform mean of 77.4 percent (Table 3). The low-birth-weight share rises by 0.67 percentage points (s.e. 0.22), or 6.8 percent of its baseline share. The very-low-birth-weight estimate is also positive, at 6.7 percent of its baseline mean, but is not statistically distinguishable from zero.[10] Infant and maternal mortality estimates are also statistically indistinguishable from zero. Duggan (2004) likewise finds no detectable effect of California's Medicaid managed-care mandates on in-hospital mortality among Medicaid-insured infants. But the fall in early prenatal care and the rise in low birth weight are consistent with reduced access or lower treatment intensity, and they could also reflect the change in eligibility or in the composition of mothers giving birth. Aizer et al. (2007) find the same direction for prenatal care and low birth weight when California moved Medicaid mothers into managed care.

6.3 Healthcare supply

The supply estimates measure slower growth. Ambulatory health care employment per 1,000 residents rose in the average county of the QCEW panel from 2.24 in 1990 to 4.83 in 2000

[10] Appendix Table A4 reports annual birth outcomes for all 78 counties.

(Table 1), so a negative estimate means that counties grew less after the reform reached them than counties it had not yet reached.

Ambulatory employment runs 0.2663 jobs per 1,000 residents below its counterfactual path (s.e. 0.1494), 9.6 percent of the pre-treatment mean of 2.773 and significant at the 10 percent level, and establishments per 1,000 residents are 0.0490 lower (s.e. 0.0185), 6.5 percent of 0.7494 (Table 4). Annual wages per worker are \$704 lower (s.e. 239), 3.4 percent of \$20,887 in constant 2025 dollars, and certified hospitals per 1,000 residents are 0.0006 lower (s.e. 0.0002), 3.4 percent of their pre-treatment mean. Figure 4 reports the event studies; the employment leads at $e = -8$ and $e = -7$ and the certified-hospital lead at $e = -6$ exclude zero (Section 5). Fewer than a third of resident workers, 29.3 percent in 2000, worked in a county of another reform cohort (Appendix Figure A4), which limits how much of the employment shortfall could reflect workers taking jobs in counties not yet treated.

Employees per establishment rise by 0.2543 (s.e. 0.1511), 7.6 percent of 3.356, an estimate significant at the 10 percent level. The ratio divides county employment by county establishments, so it rises when small establishments close even if no establishment grows, and I read it with caution.

NAICS 621 covers physicians' offices, outpatient clinics, and diagnostic services, so these estimates describe ambulatory care, which includes primary and preventive care. They do not measure hospital-based specialized or emergency care, for which certified hospitals are the only measure here. Slower employment growth together with lower wages per worker is what a fall in the demand for ambulatory labor produces in a competitive market, where demand shifting in along an upward-sloping supply curve lowers employment and wages together. The estimates therefore do not require providers to set wages, though they cannot rule it out, and wages per worker also fall when the mix of jobs shifts toward lower-paid positions. In the framework of Section 3, a shortfall of this kind is consistent with cost-reducing innovation and with labor-saving quality innovation alike. Its direction matches the lower staffing per patient at privatized U.S. public hospitals (Duggan et al., 2023).

Together with higher mortality, the results are consistent with cost containment whose harm to patients outweighed any quality gain, the case in which Proposition 1 (Appendix C) predicts higher mortality under private control, although the design does not isolate this mechanism.

6.4 Mortality by cause

This subsection decomposes the all-cause estimate into twelve named causes of death and a residual of all other deaths, each in deaths per 100,000 residents (Table 5; Figure 5). Each cause can depend on several services and on factors outside healthcare, so a cause-specific estimate does not measure the quality of one corresponding service (Appendix C).

Cardiovascular and metabolic causes show mixed effects. Diabetes mortality increases by 4.913 deaths per 100,000 (s.e. 2.409), or 9.5 percent of its pre-treatment mean, whereas hypertension mortality falls by 5.943 (s.e. 2.221), or 22.3 percent. Both conditions can benefit from regular monitoring and medication, but these opposing estimates do not support a general claim that chronic disease management improved or deteriorated. Heart-disease and cerebrovascular mortality estimates are 1.009 (s.e. 3.980) and -1.285 (s.e. 2.306), respectively; neither is statistically distinguishable from zero.

Among respiratory and infectious causes, AIDS mortality increases by 10.22 deaths per 100,000 (s.e. 2.177), or 41.9 percent. Antiretroviral treatment is linked to lower mortality, making treatment access a potential mechanism. Establishing that the reform disrupted this access, however, requires evidence on treatment receipt and continuity. The estimates for chronic pulmonary disease, pneumonia and influenza, and septicemia are positive but statistically indistinguishable from zero, leaving the evidence for these outcomes inconclusive.

Cancer and chronic liver disease show no common pattern. Cancer mortality falls by 9.277 deaths per 100,000 (s.e. 4.245), or 8.1 percent, but attributing that reduction to earlier diagnosis or improved treatment would require evidence on disease stage, treatment receipt,

and survival following diagnosis. The cirrhosis estimate is −0.644 (s.e. 1.245) and statistically indistinguishable from zero.

External causes raise the question of timely emergency care. Homicide mortality increases by 5.766 deaths per 100,000 (s.e. 1.974), or 41.8 percent, while the accident estimate is −1.533 (s.e. 2.433) and statistically indistinguishable from zero. Timely access to an appropriately equipped facility can matter for survival after severe injury, so the homicide estimate may also reflect reduced emergency capacity: with fewer hospitals and ambulatory establishments than without the reform, patients needing immediate care could have traveled farther. Testing this requires data on emergency transport and treatment.

The residual, all deaths less the twelve named causes, has the largest positive estimate, 20.60 deaths per 100,000 (s.e. 7.030), or 10.8 percent of its pre-treatment mean of 190.0. The increase suggests that the reform's adverse mortality effects extended beyond the named causes, but the residual mixes conditions with different care needs, so the estimate cannot say whether the increase was concentrated in a few conditions or spread across many. Finer cause-of-death and healthcare-use data would be needed to determine whether reduced access, treatment disruptions, or other mechanisms explain the increase.

7 Robustness

7.1 Conditional trends

Rollout timing may be related to county characteristics that affect mortality trends.[11] I therefore use the doubly robust estimator of Callaway and Sant'Anna (2021), which lets untreated mortality trends depend on baseline county characteristics: the 1990–1993 averages of population density, crime rates, unemployment rates, and government-hospital birth

[11] Appendix Table A1 shows statistically significant differences between cohorts at baseline, largest in population and population density.

counts, and the 1990 poverty rate and population share aged 65 and over (Table 6; Figure 6). Conditioning on each characteristic separately gives statistically significant estimates between 24.69 and 27.39 additional deaths per 100,000 residents, or 3.4 to 3.8 percent of the pre-treatment mean, close to the baseline effect of 26.96. Conditioning on all six together gives 35.71 (s.e. 10.96), or 4.9 percent, but that estimate rests on an effective comparison group of 3.17 counties, because 72 counties fall outside the 0.05 to 0.95 range of estimated propensities. In every panel of Figure 6, the lead at $e = -8$ excludes zero, as in the baseline (Section 5). The baseline differences between cohorts in these characteristics do not account for the estimated increase.

7.2 Sample, weights and scale

Table 7 changes the comparison group, the weights, and the scale, and Figure 7 reports the event studies. Appendix Table A6 reports the same kinds of specification for the causes of death and the birth outcomes.

Excluding San Juan, the most populous county and the last to enter, changes more than the sample. San Juan contributes no post-adoption effect to the baseline estimate, but its mortality is the counterfactual for every effect estimated in 1998 and 1999, when it is the only county not yet treated, so removing it removes every estimate for those years. Without San Juan, the estimate falls from 26.96 to -2.600 deaths per 100,000 residents (s.e. 12.65) and is not statistically significant (Table 7, column 2). In the baseline event study (Figure 2), mortality effects rise with exposure and become distinguishable from zero around the years this exclusion drops, which fits a lagged response of mortality to the reform. Of the 77 treated counties in this sample, the 65 that entered between 1994 and 1997 contribute effects, and the 12 that entered in 1998 act only as comparisons; by 1997 the reform had reached about 70 percent of the population (Figure 1, panel (c)). The exclusion therefore changes the comparison group, the treated cohorts, and the exposure years behind the aggregate

estimate.

Weighting counties by their 1990–1993 population gives more influence to the populous counties, where provider concentration and reliance on public care differ, and less to small counties, whose mortality rates are more volatile. The weighted estimate is 32.27 additional deaths per 100,000 residents (s.e. 8.408), or 4.5 percent of the pre-treatment mean, against 3.7 percent in the baseline (Table 7, column 3), so the increase keeps its sign and significance when counties count by population.

Rates can change because deaths change, because population changes, or both, which matters if the reform moved people through migration. In death counts, the estimate is 173.2 additional deaths per county-year (s.e. 9.233), or 54.2 percent of the pre-treatment mean of 319.3 (Table 7, column 4), but this number does not estimate the reform's effect on deaths. In its event study (Figure 7), deaths in counties the reform reached were already rising relative to counties not yet treated from eight years before it reached them, so the post-entry estimates combine any effect of the reform with that earlier trend. Death counts grow with a county's population, so counties whose populations grow at different rates add deaths at different rates even when their death rates move together; dividing by population removes that difference, which is why the main outcome is the rate. Whether the reform moved population is a separate question that the counts cannot answer.

In logs, the estimate is 0.0247 (s.e. 0.0145), about 2.5 percent, smaller than the baseline and significant only at the 10 percent level (Table 7, column 5).[12]

7.3 Mainland comparison

I also use mainland counties in states that had not adopted Medicaid managed care by 2000 as a comparison group. Mainland counties cannot receive spillovers from treated counties on

[12]The transformation is $\log(1 + m_{it})$, where m_{it} denotes deaths per 100,000 residents. The conversion $100[\exp(0.0247) - 1]$ equals 2.5 percent on the multiplicative scale for $1 + m_{it}$.

the island,[13] and they allow effects in 2000, including San Juan's adoption-year effect. The doubly robust estimator of Callaway and Sant'Anna (2021) lets untreated trends depend on 1990–1993 population density and the 1990 poverty rate. With 282 comparison counties, the estimate through 2000 is 23.15 additional deaths per 100,000 residents (s.e. 20.54), or 3.2 percent of the pre-treatment mean of 731.8 (Table 7, column 6), close to the baseline estimate but statistically indistinguishable from zero.[14] The same comparison fails by cause of death. Appendix Tables A8 and A10 and Appendix Figure A2 show cirrhosis mortality up 573.9 percent through 2000, an increase no healthcare reform produces. The island's and CDC WONDER's cause coding differ, every mainland horizon crosses the 1999 change from ICD-9 to ICD-10, and WONDER withholds counts of nine or fewer, so I do not use the cause-specific mainland estimates.

7.4 Alternative estimators and the two-way fixed effects decomposition

Table 8 reports four estimators that allow treatment effects to vary across cohorts and time. The baseline estimator of Callaway and Sant'Anna (2021) yields 26.96 additional deaths per 100,000 residents (s.e. 10.32), against 31.08 (s.e. 9.090) for Sun and Abraham (2021). The extended two-way fixed effects estimator of Wooldridge (2025) and the imputation estimator of Borusyak et al. (2024) both yield 37.08, with standard errors of 7.533 and 7.320. All four

[13] Within the island, spillovers would run through workers and patients who crossed between cohorts. Fewer than a third of resident workers, 29.3 percent in 2000, worked in a county of another cohort, the channel for spillovers among workers. Patients could cross cohorts to reach hospitals, which were concentrated in 22 counties, although in 1993 every cohort had at least one county with a government hospital and one with a private hospital (Appendix Figure A4). Treated residents who were cared for in hospitals that had not changed hands, including the two the Commonwealth retained, or who crowded a comparison county's hospitals, would bias the estimate toward zero. Providers who moved into comparison counties would instead lower comparison mortality and bias the estimate away from zero, which these data cannot rule out.

[14] Alternative mainland samples and estimates through 2003 are also positive and imprecise. With 353 counties in states without Medicaid health maintenance organizations (HMOs), the type of MCO Puerto Rico contracted (Centers for Medicare and Medicaid Services, 2000), the estimate through 2000 is 21.57 deaths per 100,000 (s.e. 21.67; Appendix Table A7). Through 2003, the non-MMC and non-HMO estimates are 16.22 (s.e. 19.39) and 14.93 (s.e. 27.23), or 2.2 and 2.0 percent of the same baseline mean (Appendix Table A9; Appendix Figure A3).

are statistically significant at the 1 percent level and correspond to 3.7 to 5.1 percent of the pre-treatment mean, so the baseline is the smallest of the four; Figure 8 compares their dynamic effects.

The conventional two-way fixed effects (TWFE) regression instead yields −1.283 deaths per 100,000 residents (s.e. 10.20), statistically indistinguishable from zero.[15] With staggered adoption, TWFE uses counties already treated as comparisons for counties treated later, so changes in the treatment effects of those comparison counties enter its estimate (Callaway and Sant'Anna, 2021; Borusyak et al., 2024). Following de Chaisemartin and D'Haultfœuille (2020), TWFE here places negative weight on 32.3 percent of treated county-years, and the negative weights sum to 16.3 percent of the positive ones in absolute value (Appendix Table A5). The TWFE coefficient therefore need not recover the average treatment effect, which is why the baseline uses Callaway and Sant'Anna (2021) and TWFE is a diagnostic here.

8 Conclusion

This paper examines how Puerto Rico moved its public healthcare from public to private management. The baseline estimates indicate that the reform increased all-cause mortality by 26.96 annual deaths per 100,000 residents, equivalent to 3.7 percent of the pre-treatment mean. These findings provide evidence of adverse population health effects during the transition. The estimate stays positive and significant when counties that differ at baseline may follow different trends, when counties are weighted by population, and under three other heterogeneity-robust estimators. It is smaller and significant only at the 10 percent level in logs, and it is indistinguishable from zero against mainland counties and in a sample without the last county to enter, which also removes every estimate for 1998 and 1999.

[15] The six specifications in Appendix Table A5 add different contemporaneous covariates to TWFE; their estimates range from −1.756 to +3.619, and none is statistically significant.

I examined two mechanisms, slower growth in ambulatory healthcare supply and lower use of timely care. Employment and establishments in ambulatory health care grew less in counties the reform had reached than in counties it had not yet reached, and the share of births with early prenatal care fell relative to the counterfactual. These patterns are consistent with the theoretical framework's concern that stronger incentives to reduce costs can harm patients when relevant dimensions of healthcare are difficult to specify, measure, and enforce in contracts (Section 3). Fabre and Straub (2023) find the results of the public-private partnerships (PPPs) they survey mostly disappointing and trace them in part to weak incentives for providers to improve quality, and Medicaid managed care has also been found to limit timely access to care and worsen health outcomes for some beneficiaries (Aizer et al., 2007). The framework also allows quality improvements, and the mixed cause-specific mortality estimates may indicate differences in benefits across health services.

The paper contributes evidence on the management of healthcare programs under public financing. Because the reform changed facility control, provider payment, and eligibility together, the estimates measure their joint effect. Future work could estimate the reform's fiscal savings and its net effect on welfare. Longer-run comparisons would help distinguish disruption from persistent effects; variation separating policy components would help explain which features affected patients the most.

The estimates speak most directly to publicly financed systems that still own and staff their facilities. The closest current case is the Veterans Health Administration, which buys a growing share of its care from private providers (Section 1); moving all of veterans' care to private providers is the nearest analogue to the transfer studied here, with the same difficulty of writing the quality of care into contracts. Mainland Medicaid already pays private providers, and states already contract much of its care to private managed-care organizations (Centers for Medicare and Medicaid Services, 2000), so the estimates do not predict the effect of extending managed care further. Read in reverse, they indicate what a move from private management of publicly financed care toward public provision would aim

to recover, although nothing in the design shows that undoing the transfer would reverse its effect. The transfer also came with a change in eligibility, on an island where no mainland provider was within reach for routine care, and in the 1990s, so the effect elsewhere may differ in size.

The findings nevertheless show why changes in patient health and provider capacity matter when assessing policies intended to reduce costs. For settings undertaking comparable transitions, evaluating expenditure alongside access and patient health outcomes is necessary to assess whether stronger cost-reduction incentives improve performance while preserving the benefits of publicly financed care. Before renewing managed-care contracts, governments should therefore check whether any spending reductions are accompanied by worse patient health outcomes.

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I. Tables

Table 1: Descriptive statistics by year, Puerto Rico counties

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Mortality, births and population											
All-cause mortality per 100,000	703.5	696.9	720.2	744.5	728.4	768.4	763.5	759.8	767.2	734.7	718.3
	(116.7)	(103.9)	(103.9)	(108.0)	(120.9)	(132.4)	(126.5)	(114.1)	(114.3)	(109.6)	(113.2)
Government-hospital birth share (percent)	65.8	69.2	65.3	64.5	63.5	59.6	51.6	45.2	36.6	33.6	30.6
	(11.3)	(33.4)	(10.9)	(11.2)	(10.4)	(11.4)	(15.2)	(15.2)	(14.6)	(18.6)	(19.9)
Private-hospital birth share (percent)	33.8	35.3	34.4	35.5	36.1	39.6	48.1	54.6	63.1	66.2	69.2
	(11.3)	(18.5)	(11.0)	(10.7)	(10.6)	(11.7)	(15.3)	(15.3)	(14.6)	(18.6)	(19.9)
Population	45,228	45,502	45,884	46,430	47,253	47,684	47,240	47,638	48,053	48,489	48,828
	(59,205)	(59,564)	(60,064)	(60,778)	(61,281)	(59,393)	(58,898)	(58,908)	(58,930)	(58,966)	(59,001)
Unemployment rate (percent)	17.56	17.78	18.99	19.68	18.83	16.08	16.23	15.57	16.16	14.90	13.24
	(4.56)	(4.99)	(4.95)	(5.11)	(4.94)	(4.40)	(4.57)	(4.60)	(4.70)	(4.55)	(4.15)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Panel B: Ambulatory Health Care Services (NAICS 621)											
Employment per 1,000 residents	2.24	2.60	2.75	2.90	3.20	3.27	3.71	4.18	4.47	4.68	4.83
	(2.15)	(2.96)	(2.79)	(2.70)	(3.02)	(2.78)	(2.96)	(3.32)	(3.59)	(3.47)	(3.56)
Establishments per 1,000 residents	0.670	0.705	0.765	0.773	0.782	0.828	0.887	0.944	0.990	1.065	1.089
	(0.473)	(0.479)	(0.515)	(0.513)	(0.518)	(0.564)	(0.576)	(0.601)	(0.616)	(0.640)	(0.657)
Employees per establishment	3.11	3.28	3.27	3.46	3.81	3.77	4.01	4.20	4.29	4.22	4.27
	(1.37)	(1.70)	(1.60)	(1.54)	(1.67)	(1.40)	(1.46)	(1.50)	(1.69)	(1.47)	(1.44)
Wages per worker (2025 USD)	20,104	20,134	20,821	21,283	22,031	22,796	22,762	23,512	23,702	23,848	23,742
	(5,470)	(5,170)	(5,251)	(5,107)	(4,945)	(4,843)	(4,238)	(4,324)	(4,678)	(4,145)	(4,729)
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
% of employment observed	97.9	97.8	97.8	97.9	97.7	97.0	96.9	96.8	96.4	96.5	96.8
Panel C: Medicare-certified providers											
Hospitals per 1,000 residents		0.018	0.018	0.018	0.018	0.018	0.018	0.017	0.017	0.018	0.017
		(0.026)	(0.027)	(0.027)	(0.027)	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)	(0.025)
Hospital beds per 1,000 residents		1.55	1.57	1.56	1.54	1.54	1.55	1.53	1.52	1.60	1.52
		(2.55)	(2.58)	(2.57)	(2.54)	(2.56)	(2.56)	(2.54)	(2.53)	(2.65)	(2.48)
Treated counties		0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties		78	78	78	60	33	26	13	1	1	0
% of population observed		100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Sources: Panel A, the Puerto Rico Planning Board's municipal series; Panel B, the Bureau of Labor Statistics Quarterly Census of Employment and Wages; Panel C, the Medicare Provider of Services file. Percent of employment observed is the employment of the counties this panel covers in that year over the island total for the same year; employees per establishment is county employment over county establishments; wages are annual payroll over average annual employment, in 2025 dollars; the government-hospital birth share is births in a government-operated hospital divided by all births.

Table 2: Effect of the Health Reform on Utilization Between Types of Providers

	(1) Government-hospital births	(2) Private-hospital births
ATT	-0.1794***	0.1729***
	(0.0125)	(0.0119)
Pre-treatment mean	0.6418	0.3642
ATT as % of pre-treatment mean	-28.0%	47.5%
Treated counties	78	78

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Outcomes are share of births.

Table 3: Effect of the Health Reform on Health Outcomes

	(1) All causes per 100,000	(2) Infant mortality per 1,000 births	(3) Maternal mortality per 100,000 births	(4) Low birth weight share of births	(5) Very low birth weight share of births	(6) Early prenatal care share of births
ATT	26.96***	0.0985	-4.291	0.0067***	0.0008	-0.0233***
	(10.32)	(0.7941)	(22.43)	(0.0022)	(0.0010)	(0.0062)
Pre-treatment mean	724.2	12.56	19.27	0.0987	0.0121	0.7735
ATT as % of pre-treatment mean	3.7%	0.8%	-22.3%	6.8%	6.7%	-3.0%
Treated counties	78	78	78	78	78	78

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Low birth weight, very low birth weight and early prenatal care are unavailable for 1992. Each covers all 78 counties in the years it appears. Dataset is unbalanced.

Table 4: Effect of the Health Reform on Health Supply

	(1) Employment per 1,000 residents	(2) Establishments per 1,000 residents	(3) Employees per establishment ratio	(4) Wages per worker 2025 USD	(5) Certified hospitals per 1,000 residents
ATT	-0.2663*	-0.0490***	0.2543*	-703.9***	-0.0006***
	(0.1494)	(0.0185)	(0.1511)	(238.9)	(0.0002)
Pre-treatment mean	2.773	0.7494	3.356	20,886.5	0.0166
ATT as % of pre-treatment mean	-9.6%	-6.5%	7.6%	-3.4%	-3.4%
Treated counties	59	59	59	59	78

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Columns (1) to (4) come from the Bureau of Labor Statistics Quarterly Census of Employment and Wages and column (5) from the Medicare Provider of Services file.

Table 5: Effect of the Health Reform on Mortality by Cause

Panel A. All-cause mortality				
	(1)			
	All causes			
ATT	26.96***			
	(10.32)			
Pre-treatment mean	724.2			
ATT as % of pre-treatment mean	3.7%			
Treated counties	78			
Panel B. Cardiovascular and metabolic causes				
	(2)	(3)	(4)	(5)
	Heart disease	Cerebrovascular	Hypertension	Diabetes
ATT	1.009	-1.285	-5.943***	4.913**
	(3.980)	(2.306)	(2.221)	(2.409)
Pre-treatment mean	156.6	36.72	26.68	51.55
ATT as % of pre-treatment mean	0.6%	-3.5%	-22.3%	9.5%
Treated counties	78	78	78	78
Panel C. Respiratory and infectious causes				
	(6)	(7)	(8)	(9)
	Chronic pulmonary	Pneumonia and influenza	Septicemia	AIDS
ATT	1.766	0.9216	2.097	10.22***
	(2.205)	(2.363)	(1.362)	(2.177)
Pre-treatment mean	29.12	29.44	11.31	24.39
ATT as % of pre-treatment mean	6.1%	3.1%	18.5%	41.9%
Treated counties	78	78	78	78
Panel D. Cancer and chronic liver disease				
	(10)	(11)		
	Cancer	Cirrhosis		
ATT	-9.277**	-0.6440		
	(4.245)	(1.245)		
Pre-treatment mean	114.4	18.64		
ATT as % of pre-treatment mean	-8.1%	-3.5%		
Treated counties	78	78		
Panel E. External causes				
	(12)	(13)		
	Accidents	Homicide		
ATT	-1.533	5.766***		
	(2.433)	(1.974)		
Pre-treatment mean	36.02	13.78		
ATT as % of pre-treatment mean	-4.3%	41.8%		
Treated counties	78	78		
Panel F. Unclassified				
	(14)			
	Unclassified			
ATT	20.60***			
	(7.030)			
Pre-treatment mean	190.0			
ATT as % of pre-treatment mean	10.8%			
Treated counties	78			

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Chronic pulmonary and homicide are unavailable for 1995; septicemia is unavailable for 1990, 1991, 1992, 1993 and 1995; AIDS is unavailable for 1990, 1991 and 1995. Each covers all 78 counties in the years it appears. Unclassified is all-cause deaths less the causes printed above it, so it closes the decomposition rather than naming a further cause. Each cause in the table is the following share of the pre-treatment all-cause mean: heart disease 22%, cerebrovascular 5%, hypertension 4%, diabetes 7%, chronic pulmonary 4%, pneumonia and influenza 4%, septicemia 2%, AIDS 3%, cancer 16%, cirrhosis 3%, accidents 5%, homicide 2%, unclassified 26%. Outcomes are per 100,000. Dataset is unbalanced.

Table 6: Alternative Specifications (with Conditional Trends)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All-cause mortality								
ATT (per 100,000)	26.96***	25.52**	26.84**	26.52**	24.69**	26.01**	27.39***	35.71***
	(10.32)	(10.69)	(10.44)	(10.53)	(10.19)	(10.88)	(10.41)	(10.96)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	3.7%	3.5%	3.7%	3.7%	3.4%	3.6%	3.8%	4.9%
Covariates								
Population density		✓						✓
Age 65 and over share			✓					✓
Crime				✓				✓
Poverty rate					✓			✓
Unemployment rate						✓		✓
Pre-reform government births							✓	✓
Treated counties	78	78	78	78	78	78	78	78
Estimated through	1999	1999	1999	1999	1999	1999	1999	1999
Effective comparison counties	.	20.28	25.28	21.35	8.659	12.17	19.49	3.173
Group-time cells dropped	0	0	0	0	0	0	0	0

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Each control variable is one number per county. 1990 to 1993 means: population density, crime, unemployment rate and pre-reform government births. 1990 Census values: age 65 and over share and poverty rate. Density is residents per square mile; the age 65 and over share is the 1990 Census counts aged 65 to 84 and 85 and over, over total county population; crime is total reported offences per 1,000 residents, murders included; the poverty rate is the fraction of families below the Census poverty threshold, of all families; unemployment is the county annual rate; pre-reform government births is a count of births in government-operated hospitals and not a share of births. Conditional columns reweight the comparison group by the estimated odds of cohort membership at each cohort's base period; 72 counties fall outside the 0.05 to 0.95 range of that probability in column (8), where the weights are least even.

Table 7: Alternative Specifications

	(1) Baseline	(2) Excluding San Juan	(3) Population weighted	(4) Levels	(5) Logs	(6) Never-treated
All-cause mortality						
ATT	26.96***	-2.600	32.27***	173.2***	0.0247*	23.15
	(10.32)	(12.65)	(8.408)	(9.233)	(0.0145)	(20.54)
Pre-treatment mean	724.2	724.2	724.2	319.3	6.576	731.8
ATT as % of pre-treatment mean	3.7%	-0.4%	4.5%	54.2%	2.5%	3.2%
Treated counties	78	77	78	78	78	78
Never-treated counties						282

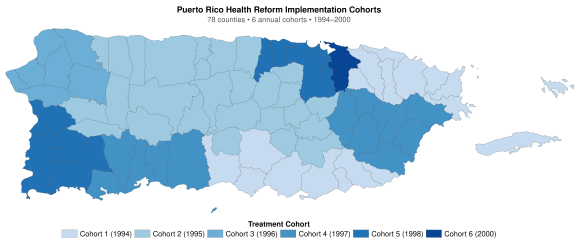
Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. The outcome is measured per 100,000. Every column is the Callaway and Sant'Anna estimator. The never-treated column is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), the same covariates as the never-treated tables. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10).

Table 8: Alternative Estimators

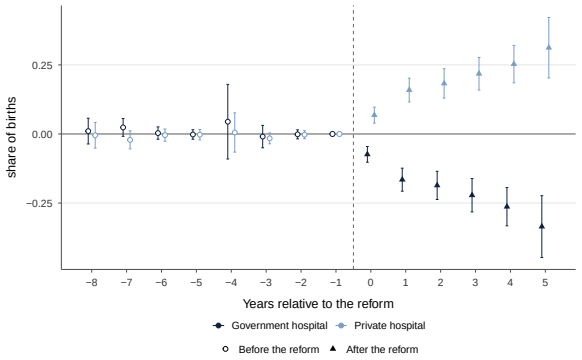
	(1) Callaway & Sant'Anna	(2) TWFE	(3) Sun & Abraham	(4) ETWFE	(5) Borusyak, Jaravel & Spiess
All-cause mortality					
ATT	26.96***	-1.283	31.08***	37.08***	37.08***
	(10.32)	(10.20)	(9.090)	(7.533)	(7.320)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2
ATT as % of pre-treatment mean	3.7%	-0.2%	4.3%	5.1%	5.1%
Treated counties	78	78	78	78	78

Notes: TWFE is two-way fixed effects OLS and ETWFE is the extended two-way fixed effects estimator of Wooldridge (2025). The named columns are Callaway and Sant'Anna (2021), Sun and Abraham (2021), and the imputation estimator of Borusyak, Jaravel and Spiess (2024). Callaway and Sant'Anna is the baseline identification strategy. Standard errors in parentheses: for CS, county-clustered multiplier bootstrap, 1,000 replications; for TWFE, SA, ETWFE and BJS, clustered on the county. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. For Callaway and Sant'Anna, the control group is the not-yet-treated counties. The outcome is measured per 100,000.

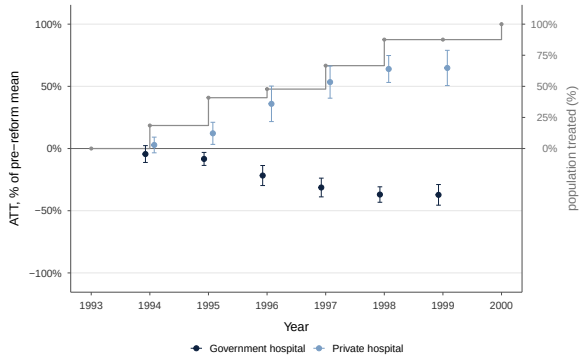
II. Figures



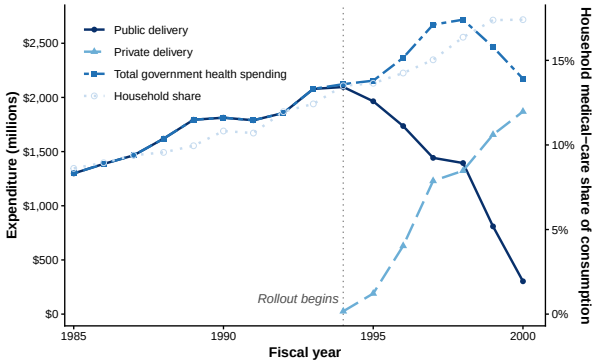
(a) Adoption cohorts



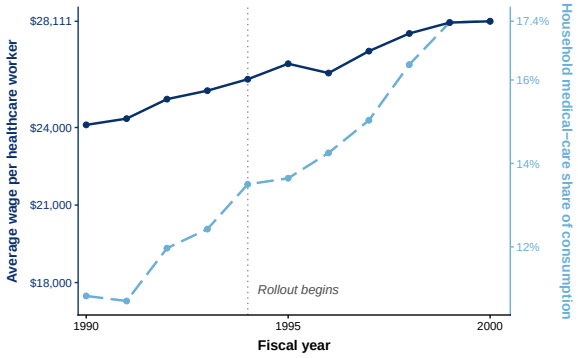
(b) Hospital birth share by ownership



(c) First stage and the rollout, by calendar year



(d) Delivery spending and managed-care purchasing



(e) Household medical spending and real wages

Figure 1: Setting

The event-study and calendar-year panels report Callaway and Sant'Anna (2021) estimates against not-yet-treated counties, Puerto Rico counties, 1990 to 1999; in the event-study panel the year before adoption is the reference period and is set to zero. Dollar series are in constant 2025 dollars. The wage is the QCEW NAICS 621 wage per worker, by calendar year. The financing appendix defines each spending series and its source. In panels (b) and (c), bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

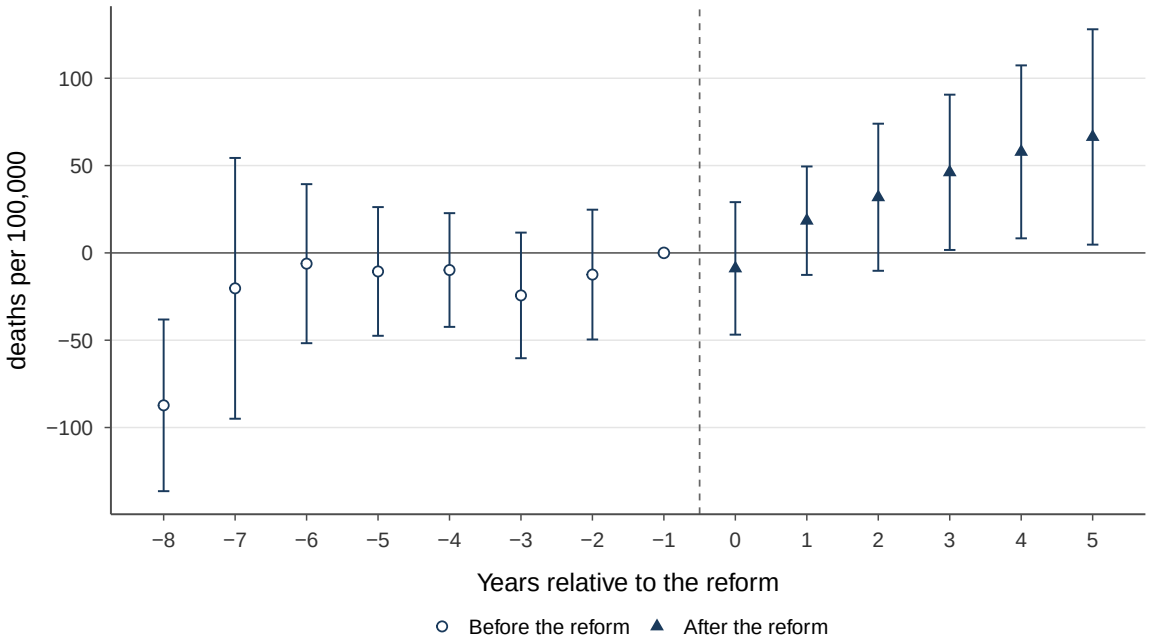
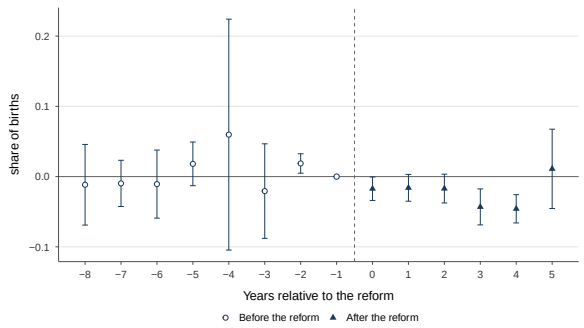
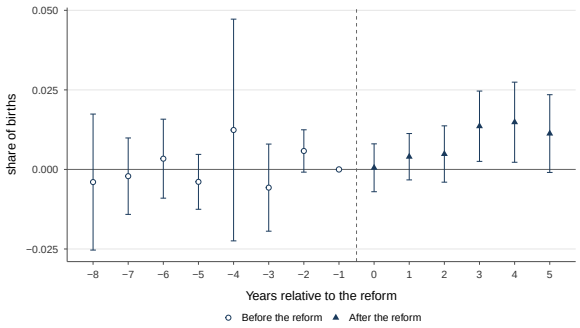


Figure 2: All-cause mortality

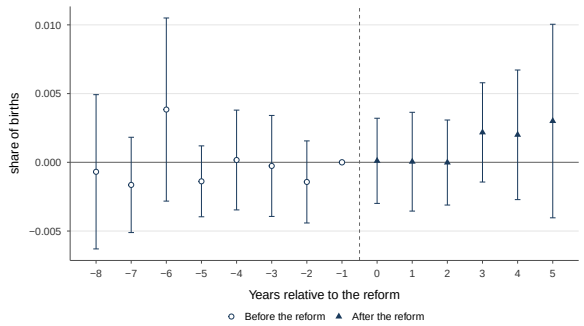
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



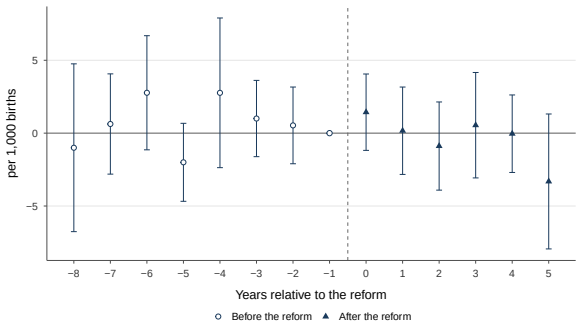
(a) Early prenatal care



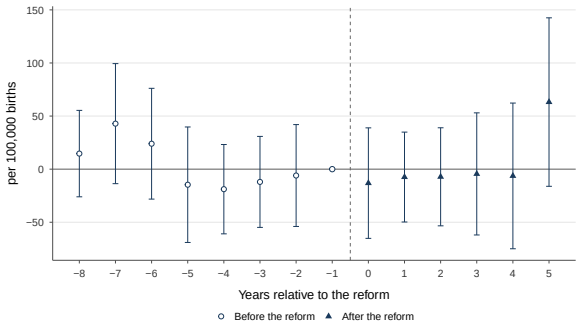
(b) Low birth weight



(c) Very low birth weight



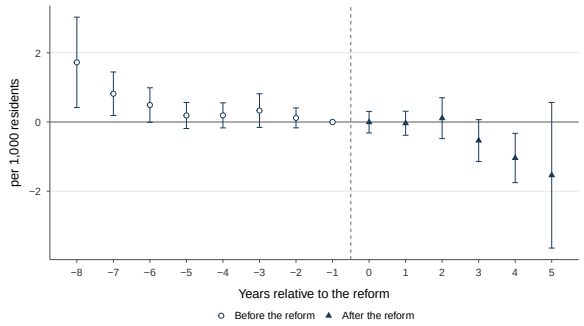
(d) Infant mortality



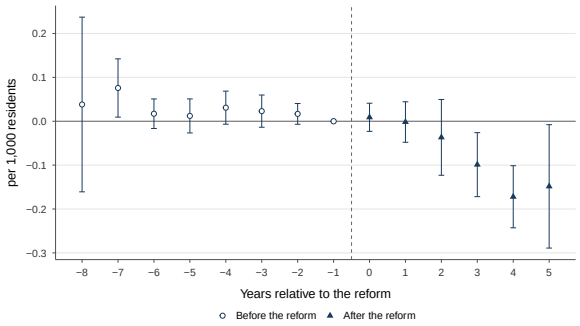
(e) Maternal mortality

Figure 3: Prenatal care and prenatal-care-sensitive outcomes

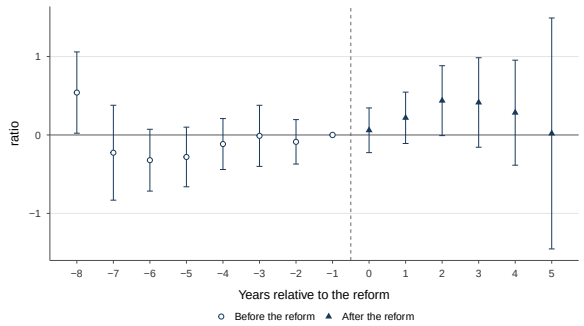
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



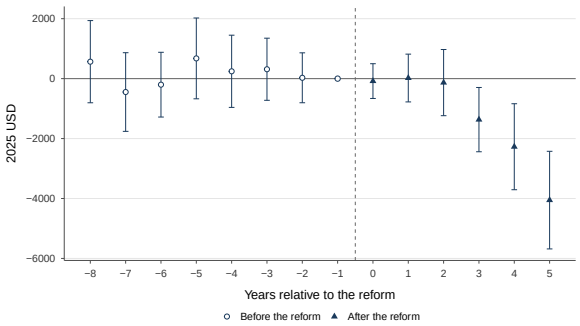
(a) Employment



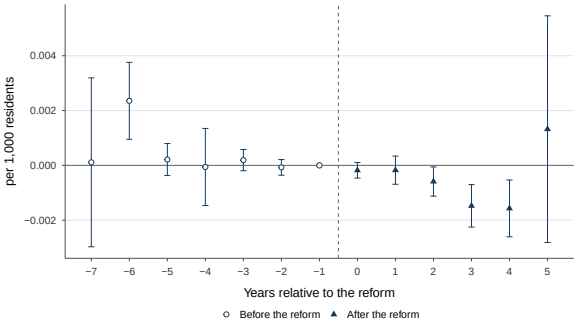
(b) Establishments



(c) Employees per establishment



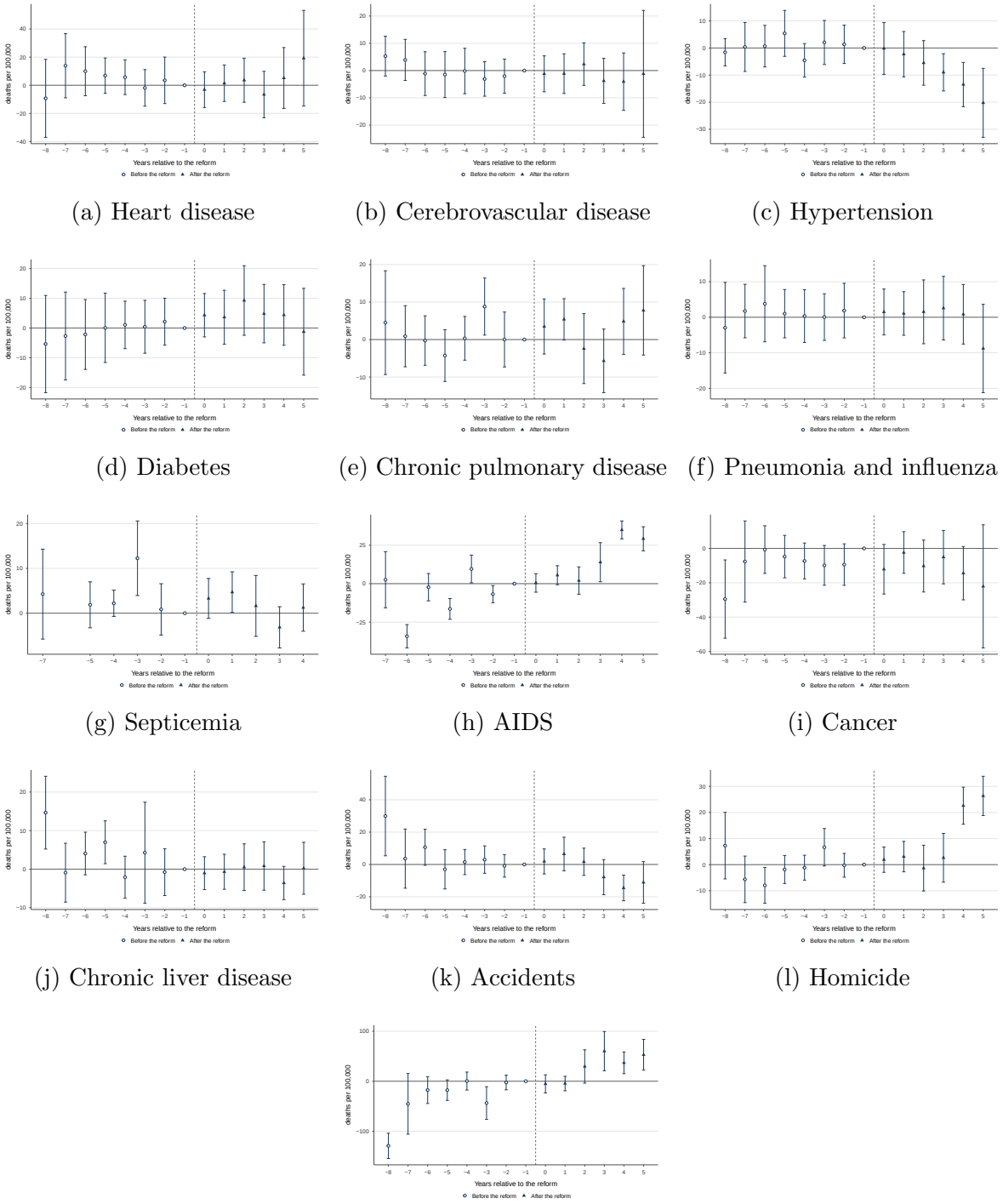
(d) Wages per worker



(e) Certified hospitals

Figure 4: Health supply

Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Employment, establishments, employees per establishment and wages are QCEW NAICS 621, ambulatory health care services, for the counties with published figures in every year, 1990 to 1999. Certified hospitals come from the Medicare Provider of Services file, 1991 to 1999, for every county. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



(a) Heart disease (b) Cerebrovascular disease (c) Hypertension

(d) Diabetes (e) Chronic pulmonary disease (f) Pneumonia and influenza

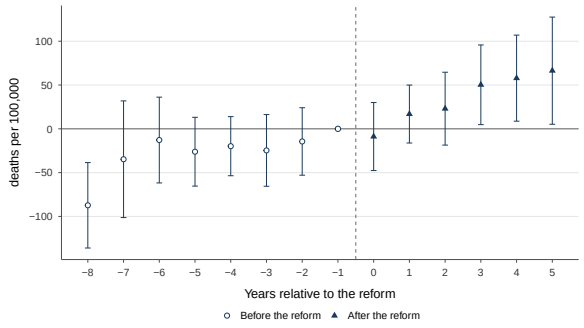
(g) Septicemia (h) AIDS (i) Cancer

(j) Chronic liver disease (k) Accidents (l) Homicide

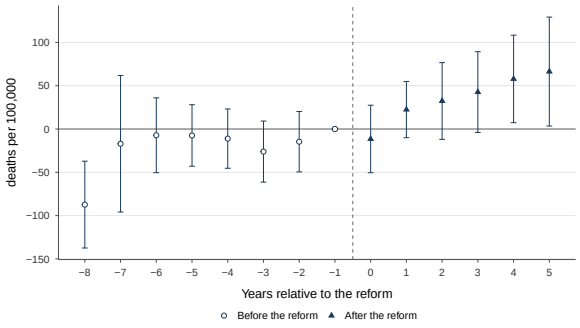
(m) Unclassified

Figure 5: Mortality by cause

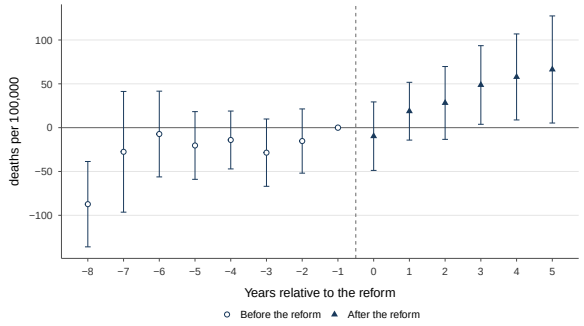
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Unclassified is all-cause deaths less the causes in the other panels. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



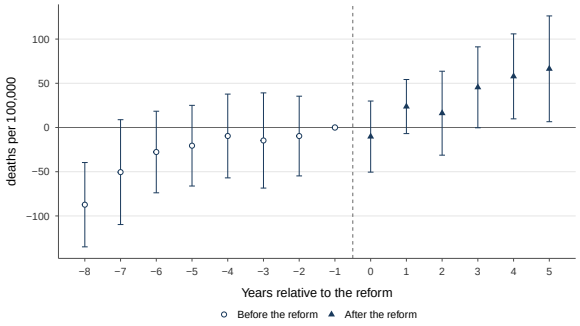
(a) Population density



(b) Age 65 and over share

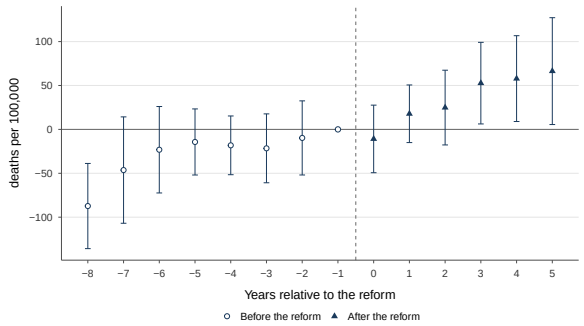


(c) Crime

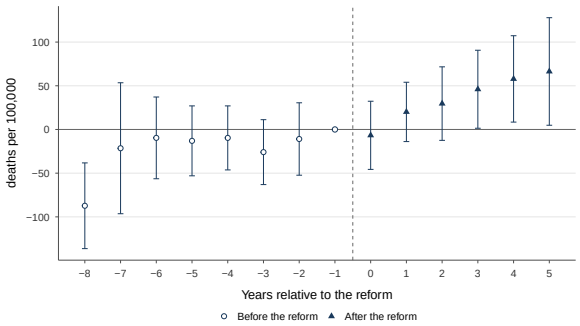


(d) Poverty rate

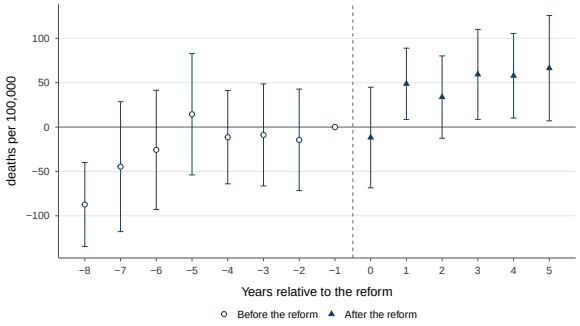
Figure 6: All-cause mortality under conditional trends



(e) Unemployment rate



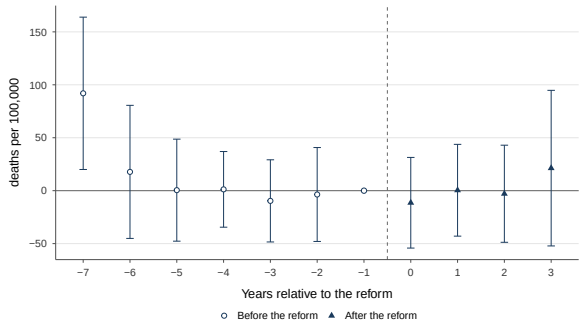
(f) Pre-reform government births



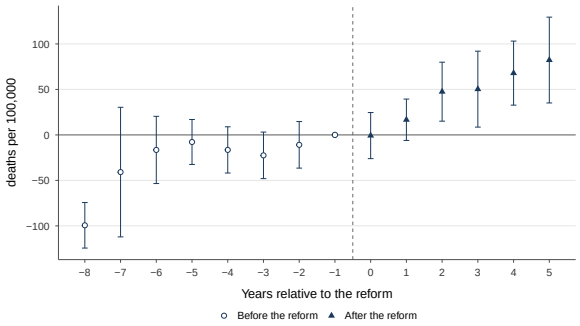
(g) All six covariates

Figure 6: All-cause mortality under conditional trends (continued)

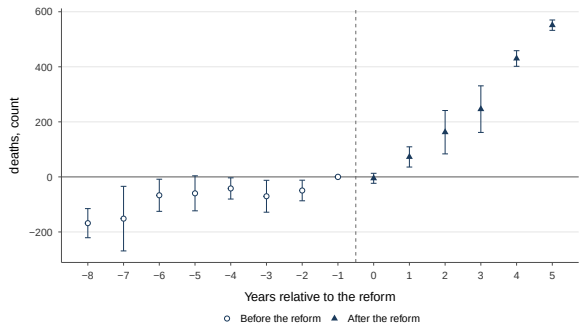
Callaway and Sant'Anna (2021) doubly robust estimates by year relative to adoption, conditioning on the baseline covariates each panel names: 1990 to 1993 means of population density, crime, the unemployment rate and pre-reform government births, and 1990 Census values of the age 65 and over share and the poverty rate. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



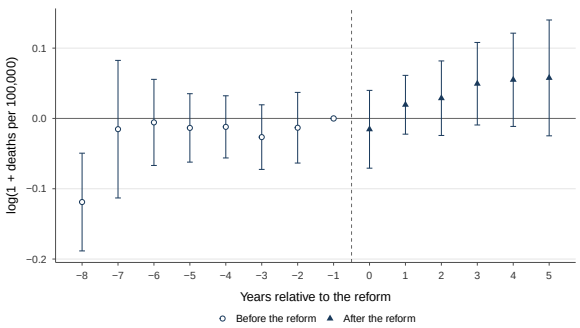
(a) Excluding San Juan



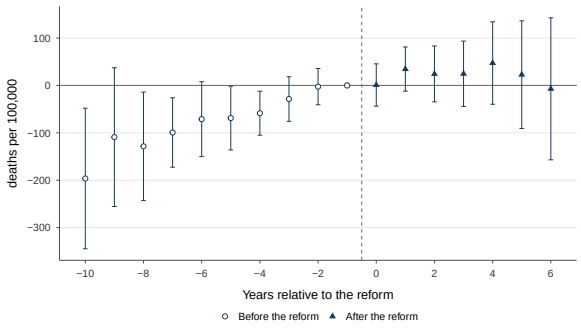
(b) Population weighted



(c) Levels



(d) Logs



(e) Never-treated, conditional

Figure 7: All-cause mortality under alternative specifications

Callaway and Sant'Anna (2021) estimates by year relative to adoption, one specification per panel. Outside the never-treated panel, the control group is the not-yet-treated Puerto Rico counties, 1990 to 1999, and the population-weighted panel weights counties by their 1990 to 1993 mean population. The never-treated panel compares Puerto Rico counties with never-treated mainland counties in states that had not adopted Medicaid managed care by 2000, conditional on population density (1990 to 1993 mean) and the 1990 Census poverty rate, through 2000. The year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

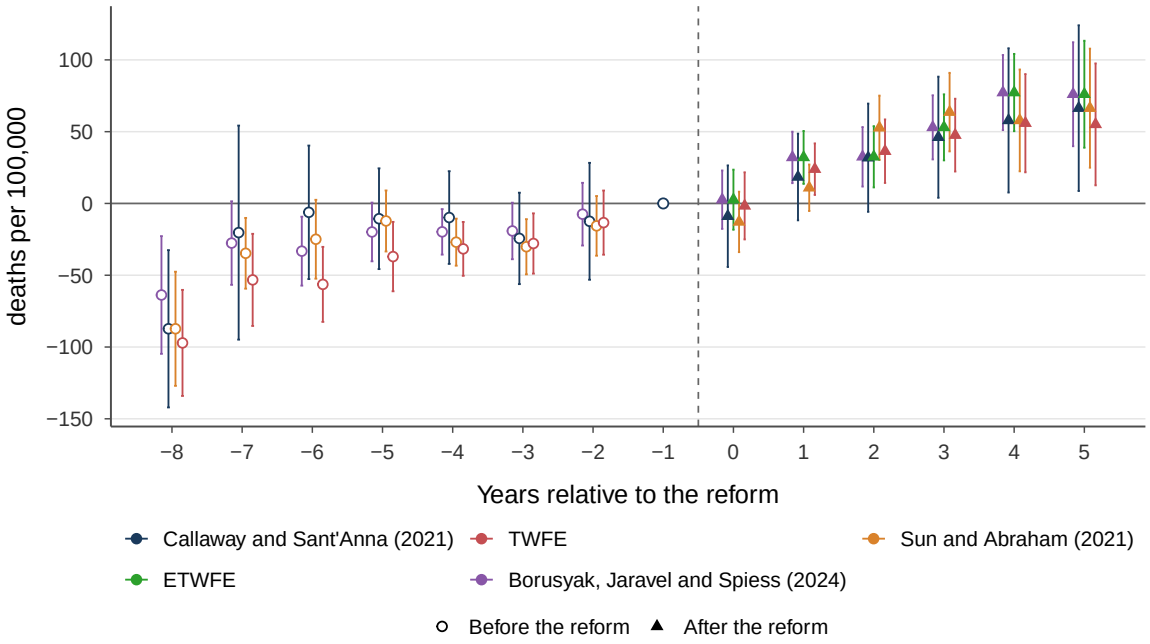


Figure 8: Alternative estimators

Event-study estimates by year relative to adoption, one series per estimator. TWFE is two-way fixed effects and ETWFE is the extended two-way fixed effects estimator of Wooldridge (2025), which estimates effects from adoption onward only. For Callaway and Sant'Anna (2021), the control group is the not-yet-treated counties and the year before adoption is the reference period, set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications for Callaway and Sant'Anna (2021), and 95 percent pointwise confidence intervals from standard errors clustered on the county for the other series.

A Additional tables and figures

Tables

Table A1: Pre-reform means by treatment cohort

Variable	1994	1995	1996	1997	1998	2000	F
Ambulatory health care employment per 1,000 residents	3.07	2.12	1.93	2.58	2.99	10.04	2.17
Ambulatory health care establishments per 1,000 residents	0.643	0.608	0.678	0.823	0.826	2.494	3.80
Ambulatory health care employees per establishment	4.17	3.27	2.70	2.73	3.25	4.03	1.18
Ambulatory health care wages per worker (2025 USD)	22,774	20,723	18,055	19,805	19,842	26,369	0.92
Hospitals per 1,000 residents	0.027	0.013	0.013	0.021	0.012	0.045	1.02
Hospital beds per 1,000 residents	1.58	1.06	0.91	2.27	1.67	8.73	2.22
All-cause mortality per 100,000	725.5	678.0	711.6	745.6	730.2	1,065.7	4.61
Government-hospital birth share (percent)	74.5	67.1	59.1	65.3	57.2	60.2	3.10
Private-hospital birth share (percent)	25.3	36.3	39.7	34.4	42.6	37.0	5.24
Population	36,717	29,535	35,263	51,767	62,300	443,627	23.53
Unemployment rate (percent)	19.43	18.15	21.54	20.37	14.80	10.17	3.97
Population per square mile	1,021.2	714.1	896.3	940.0	2,268.8	9,280.9	16.53
PAN recipients per 1,000 residents	455.6	529.2	458.3	466.4	358.7	313.1	7.15
Population aged 65 and over, 1990 (percent)	9.0	8.9	9.4	9.3	9.7	13.5	2.71
Families below the poverty threshold, 1990 (percent)	5,821.1	6,603.7	6,414.3	6,138.5	5,166.7	4,400.0	6.43
Labor force per 1,000 residents	305.7	287.9	345.6	301.9	340.8	352.9	4.24

Notes: Cells are county means over 1990–1993, averaged within treatment cohort. The 5 percent critical value at the median degrees of freedom is 2.34.

Table A2: Ambulatory Health Care Services employment and establishments by ownership sector, Puerto Rico (NAICS 621)

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: County Ambulatory Health Care Services (QCEW)											
Employment per 1,000 residents	2.24	2.60	2.75	2.90	3.20	3.27	3.71	4.18	4.47	4.68	4.83
	(2.15)	(2.96)	(2.79)	(2.70)	(3.02)	(2.78)	(2.96)	(3.32)	(3.59)	(3.47)	(3.56)
Establishments per 1,000 residents	0.670	0.705	0.765	0.773	0.782	0.828	0.887	0.944	0.990	1.065	1.089
	(0.473)	(0.479)	(0.515)	(0.513)	(0.518)	(0.564)	(0.576)	(0.601)	(0.616)	(0.640)	(0.657)
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
Panel B: County Ambulatory Health Care Services by ownership (QCEW)											
Employment	209.5	228.3	242.6	262.4	285.2	289.7	317.4	346.2	374.3	390.9	407.2
	(583.0)	(603.4)	(629.9)	(689.9)	(742.9)	(729.1)	(770.3)	(784.7)	(856.3)	(873.1)	(901.3)
Total	12,362	13,472	14,313	15,484	16,827	17,091	18,726	20,426	22,084	23,064	24,025
Government total											
Private total	12,362	13,472	14,313	15,484	16,827	17,091	18,726	20,426	22,084	23,064	24,025
Establishments	56.86	58.54	63.44	64.41	65.42	68.14	71.36	75.68	79.32	85.25	87.34
	(149.65)	(149.07)	(158.73)	(158.62)	(157.91)	(158.63)	(162.51)	(169.88)	(174.00)	(183.00)	(184.59)
Total	3,355	3,454	3,743	3,800	3,860	4,020	4,210	4,465	4,680	5,030	5,153
Government total											
Private total	3,355	3,454	3,743	3,800	3,860	4,020	4,210	4,465	4,680	5,030	5,153
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
% of employment observed	97.9	97.8	97.8	97.9	97.7	97.0	96.9	96.8	96.4	96.5	96.8
% of establishments observed	97.7	97.5	97.3	97.2	97.2	96.7	96.4	96.3	96.2	96.2	96.3
Panel C: State Ambulatory Health Care Services (QCEW)											
Employment per 1,000 residents	3.58	3.88	4.09	4.37	4.67	4.74	5.24	5.68	6.11	6.32	6.52
Employment, total	12,627	13,773	14,631	15,822	17,230	17,617	19,317	21,108	22,897	23,909	24,815
Establishments per 1,000 residents	0.973	0.998	1.075	1.080	1.078	1.118	1.185	1.248	1.298	1.382	1.404
Establishments, total	3,433	3,542	3,848	3,911	3,973	4,157	4,367	4,636	4,866	5,228	5,349
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Panel D: County private healthcare sector (CBP)											
Employment per 1,000 residents	12.62	10.49	11.50	11.89	12.43	12.83	14.30	15.31	16.17	17.78	18.69
	(8.79)	(8.65)	(7.98)	(9.09)	(9.38)	(9.35)	(9.05)	(9.28)	(10.00)	(11.25)	(11.27)
Employment	1,754.3	1,732.9	1,518.6	1,512.0	1,616.3	1,597.0	1,711.9	1,833.3	1,912.0	2,042.1	2,144.5
	(2,551.5)	(2,666.1)	(2,484.9)	(2,496.0)	(2,712.4)	(2,567.8)	(2,940.5)	(3,082.3)	(3,077.7)	(3,228.8)	(3,256.3)
Private total	24,560	24,261	27,335	28,728	30,710	30,343	35,949	38,499	40,153	42,885	45,035
Establishments per 1,000 residents	0.813	0.759	0.799	0.846	0.878	0.932	1.308	1.408	1.445	1.431	1.474
	(0.523)	(0.517)	(0.528)	(0.517)	(0.534)	(0.552)	(0.740)	(0.765)	(0.744)	(0.772)	(0.769)
Establishments	111.43	107.81	112.86	116.19	119.52	121.86	167.00	177.86	180.52	180.33	185.29
	(210.12)	(203.66)	(208.67)	(203.77)	(205.67)	(205.05)	(273.55)	(286.40)	(282.67)	(280.85)	(286.01)
Private total	2,340	2,264	2,370	2,440	2,510	2,559	3,507	3,735	3,791	3,787	3,891
Treated counties	0	0	0	0	4	10	12	15	20	20	21
Not-yet-treated counties	21	21	21	21	17	11	9	6	1	1	0
% of population observed	56.8	56.8	56.8	56.8	56.6	55.6	55.5	55.3	55.1	54.9	54.7
Panel E: Individual healthcare employment (Census)											
Employment per 1,000 residents	4.66										7.33
Employment, total	16,449										27,916
% of employees with no county identified	100.0										100.0
% of population observed	100.0										100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Percent of employment observed is the covered counties' employment over the island total that year, and percent of establishments observed is the same quotient on establishments; health workers are employed civilians aged 16 and over in the ambulatory categories that the data-construction appendix lists. They include the self-employed. Panel D covers counties where CBP published a figure, which it does above 50 employees in 1990–1992 and 100 in 1993–2000. SIC 80 includes hospitals and nursing and personal care; the data-construction appendix lists its components.

Table A3: Deaths by cause and share of all deaths by year, Puerto Rico

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)											
Deaths, with share of all deaths beneath											
All causes	26,148	26,328	27,397	28,494	28,444	30,196	29,871	29,119	29,990	29,145	28,550
	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)
Diabetes	1,664	1,665	1,830	1,870	1,866	2,210	2,091	2,097	2,295	2,184	2,273
	(6.36%)	(6.32%)	(6.68%)	(6.56%)	(6.56%)	(7.32%)	(7.00%)	(7.20%)	(7.65%)	(7.49%)	(7.96%)
Cancer	4,063	4,278	4,362	4,405	4,271	4,619	4,563	4,506	4,658	4,429	4,693
	(15.54%)	(16.25%)	(15.92%)	(15.46%)	(15.02%)	(15.30%)	(15.28%)	(15.47%)	(15.53%)	(15.20%)	(16.44%)
Heart disease	5,910	5,635	5,812	5,845	5,783	5,775	5,925	5,889	6,038	5,364	5,222
	(22.60%)	(21.40%)	(21.21%)	(20.51%)	(20.33%)	(19.13%)	(19.84%)	(20.22%)	(20.13%)	(18.40%)	(18.29%)
Cerebrovascular	1,186	1,207	1,275	1,434	1,419	1,469	1,465	1,380	1,420	1,266	1,166
	(4.54%)	(4.58%)	(4.65%)	(5.03%)	(4.99%)	(4.86%)	(4.90%)	(4.74%)	(4.73%)	(4.34%)	(4.08%)
Hypertension	741	948	900	916	899	1,074	1,084	1,190	1,395	1,617	1,575
	(2.83%)	(3.60%)	(3.29%)	(3.21%)	(3.16%)	(3.56%)	(3.63%)	(4.09%)	(4.65%)	(5.55%)	(5.52%)
Cirrhosis	290	813	804	762	648	715	670	712	672	878	820
	(1.11%)	(3.09%)	(2.93%)	(2.67%)	(2.28%)	(2.37%)	(2.24%)	(2.45%)	(2.24%)	(3.01%)	(2.87%)
Pneumonia and influenza	1,178	1,113	1,192	1,168	1,177	1,348	1,174	1,156	1,314	1,069	785
	(4.51%)	(4.23%)	(4.35%)	(4.10%)	(4.14%)	(4.46%)	(3.93%)	(3.97%)	(4.38%)	(3.67%)	(2.75%)
Septicemia	0				499	482	634	602	599	726	601
	(0.00%)	(–)	(–)	(–)	(1.75%)	(1.63%)	(2.12%)	(2.07%)	(2.00%)	(2.49%)	(2.11%)
Chronic pulmonary	875	893	998	1,157	1,183	1,145	1,248	1,242	1,385	1,240	1,144
	(3.35%)	(3.39%)	(3.64%)	(4.06%)	(4.16%)	(3.79%)	(4.18%)	(4.27%)	(4.62%)	(4.25%)	(4.01%)
AIDS	0		1,407	1,415	1,530	1,037	1,474	833	582	610	659
	(0.00%)	(–)	(5.14%)	(4.97%)	(5.38%)	(3.50%)	(4.93%)	(2.86%)	(1.94%)	(2.09%)	(2.31%)
Homicide	568	792	838	947	998	676	928	881	819	705	700
	(2.17%)	(3.01%)	(3.06%)	(3.32%)	(3.51%)	(2.34%)	(3.11%)	(3.03%)	(2.73%)	(2.42%)	(2.45%)
Accidents	1,503	1,123	1,171	1,195	1,287	1,498	1,520	1,581	1,387	1,269	1,321
	(5.75%)	(4.27%)	(4.27%)	(4.19%)	(4.52%)	(4.96%)	(5.09%)	(5.43%)	(4.62%)	(4.35%)	(4.63%)
Unclassified	8,170	7,861	6,823	7,380	6,884	8,151	7,095	7,050	7,426	7,788	7,591
	(31.25%)	(29.86%)	(24.90%)	(25.90%)	(24.20%)	(26.99%)	(23.75%)	(24.21%)	(24.76%)	(26.72%)	(26.59%)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: Entries are TOTAL deaths in Puerto Rico, with that cause's share of all deaths in parentheses beneath. Both are island figures, not county averages. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified is all-cause deaths less the causes listed, so the shares sum to 100; where a cause is unreported its deaths fall into that row; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative. Sample: unbalanced, every county the panel observes, with no balance restriction.

Table A4: Mortality and birth-margin outcomes by year, Puerto Rico counties

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)											
Mortality											
All-cause mortality per 100,000	703.5	696.9	720.2	744.5	728.4	768.4	763.5	759.8	767.2	734.7	718.3
	(116.7)	(103.9)	(103.9)	(108.0)	(120.9)	(132.4)	(126.5)	(114.1)	(114.3)	(109.6)	(113.2)
Diabetes mortality per 100,000	46.9	46.0	54.3	52.2	50.7	59.9	57.0	57.6	62.6	58.6	59.5
	(23.4)	(17.4)	(22.0)	(23.6)	(19.2)	(22.8)	(24.1)	(21.1)	(23.2)	(19.9)	(18.9)
Cancer mortality per 100,000	108.1	112.2	116.7	118.8	111.2	118.6	118.1	119.9	120.0	108.3	117.7
	(27.6)	(27.7)	(25.0)	(27.4)	(34.6)	(27.2)	(29.8)	(29.9)	(24.1)	(30.5)	(23.8)
Heart disease mortality per 100,000	163.9	155.1	156.8	154.7	151.7	153.4	153.8	153.7	153.9	139.0	133.3
	(36.6)	(33.6)	(38.8)	(30.0)	(31.6)	(41.0)	(37.4)	(34.3)	(40.7)	(43.2)	(38.4)
Cerebrovascular mortality per 100,000	34.7	34.8	36.0	39.5	37.7	40.0	39.5	37.3	36.0	34.5	30.4
	(18.8)	(12.4)	(12.5)	(11.5)	(15.8)	(13.2)	(12.6)	(15.4)	(13.8)	(17.3)	(12.4)
Hypertension mortality per 100,000	21.2	29.3	26.7	26.0	25.7	32.9	30.4	33.6	36.6	40.5	40.0
	(9.3)	(18.9)	(11.7)	(16.8)	(12.2)	(27.9)	(14.2)	(11.7)	(12.1)	(15.1)	(13.3)
Cirrhosis mortality per 100,000	7.6	23.6	22.9	21.6	17.3	18.4	19.2	19.5	18.0	23.3	20.7
	(11.2)	(11.5)	(31.5)	(9.6)	(7.8)	(8.2)	(10.8)	(11.8)	(10.2)	(9.9)	(10.5)
Pneumonia and influenza mortality per 100,000	29.8	28.2	29.4	30.0	28.1	31.2	28.7	28.2	32.5	25.0	20.1
	(14.9)	(12.5)	(12.8)	(13.2)	(14.4)	(11.2)	(11.3)	(12.4)	(15.5)	(10.9)	(10.3)
Septicemia mortality per 100,000	0.0				12.4	14.8	16.2	15.5	14.9	17.6	14.3
	(0.0)	(–)	(–)	(–)	(7.4)	(7.3)	(9.1)	(9.1)	(7.6)	(8.3)	(7.2)
Chronic pulmonary mortality per 100,000	23.9	25.9	29.0	32.1	33.7	36.3	34.0	32.2	37.3	34.4	31.2
	(12.5)	(12.8)	(14.8)	(16.6)	(14.5)	(18.3)	(14.4)	(15.8)	(16.2)	(16.5)	(15.9)
AIDS mortality per 100,000	0.0		26.6	27.5	29.4	27.7	30.2	17.3	12.0	12.9	13.4
	(0.0)	(–)	(16.7)	(17.4)	(19.0)	(15.4)	(17.0)	(12.2)	(9.4)	(9.1)	(9.2)
Homicide mortality per 100,000	9.7	13.3	13.0	16.1	17.6	18.0	15.9	17.7	15.5	12.9	14.3
	(9.7)	(13.7)	(13.4)	(15.1)	(18.1)	(14.7)	(15.1)	(15.4)	(16.1)	(11.1)	(13.2)
Accident mortality per 100,000	43.3	32.8	33.7	32.7	35.4	41.5	39.4	40.8	37.6	30.9	31.7
	(22.7)	(15.2)	(13.7)	(12.5)	(15.7)	(23.7)	(14.2)	(15.1)	(15.9)	(13.1)	(12.3)
Unclassified mortality per 100,000	214.5	195.7	175.7	193.3	177.3	181.1	181.0	186.5	190.3	196.9	191.7
	(61.6)	(43.3)	(40.9)	(39.1)	(37.1)	(56.1)	(38.0)	(39.0)	(38.8)	(38.5)	(44.2)
Birth outcomes											
Infant mortality per 1,000 live births	12.75	13.78	12.39	13.13	11.55	14.08	9.91	12.11	10.48	10.10	9.91
	(6.09)	(11.74)	(4.85)	(5.80)	(5.26)	(7.44)	(4.90)	(6.42)	(5.02)	(4.93)	(5.60)
Low birth weight share (percent)	9.28	9.98		9.65	9.86	10.20	10.49	11.07	10.98	11.36	2.46
	(1.62)	(7.51)	(–)	(1.64)	(1.44)	(1.76)	(1.90)	(2.33)	(2.24)	(2.46)	(1.34)
Early prenatal care share (percent)	71.9	77.8		77.7	78.5	77.9	78.6	79.0	79.7	79.7	78.6
	(6.7)	(34.1)	(–)	(6.7)	(7.2)	(6.2)	(5.9)	(6.8)	(6.5)	(6.4)	(8.7)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified mortality is all-cause deaths less every cause identified, expressed per 100,000 residents; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative.

Table A5: Two-Way Fixed Effects with Control Variables: Contemporaneous Covariates

	(1)	(2)	(3)	(4)	(5)	(6)
All-cause mortality						
Contemporaneous covariates, in levels						
ATT (per 100,000)	-1.283	0.7695	-1.549	-1.756	3.619	3.413
	(10.20)	(10.34)	(10.15)	(9.907)	(10.44)	(9.836)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	-0.2%	0.1%	-0.2%	-0.2%	0.5%	0.5%
SE relative to unconditional		1.013	0.9945	0.9709	1.024	0.9640
Counties	78	78	78	78	78	78
Covariates						
Population density		✓				✓
Unemployment rate			✓			✓
Crime				✓		✓
PAN recipient share					✓	✓
Estimated through	1999	1999	1999	1999	1999	1999
Parameters	.	2	2	2	2	5
Clusters	78	78	78	78	78	78

Notes: Two-way fixed effects, ordinary least squares. Each covariate enters in levels at its value in that year. Standard errors in parentheses: clustered on the county. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. Each covariate is measured in the same year as the outcome, so in a treated county it is measured after that county's reform year. Each control variable is one number per county and year. Annual county values: population density, unemployment rate, crime and PAN recipient share. The columns differ in what they condition on and rest on the same counties and years. Two-way fixed effects under staggered adoption weights 32.3% of treated county-years negatively; the negative weights are 16.3% of the positive ones in absolute value. Density is residents per square mile in that year; unemployment is the county annual rate in that year; crime is total reported offences per 1,000 residents in that year, murders included, on the definition the baseline crime control also uses; the PAN recipient share is residents receiving the Programa de Asistencia Nutricional over county population in that year.

Table A6: Alternative Specifications: Mortality by Cause and Prenatal-Care-Sensitive Outcomes

	(1) Baseline	(2) Excluding San Juan	(3) Population weighted	(4) Levels	(5) Logs	(6) Never-treated
Panel A. Mortality						
All-cause mortality						
All causes (per 100,000; pre-treatment mean 724.2)	26.96***	-2.600	32.27***	173.2***	0.0247*	23.15
	(10.32)	(12.65)	(8.408)	(9.233)	(0.0145)	(20.54)
% of pre-treatment mean	3.7%	-0.4%	4.5%	54.2%	2.5%	3.2%
Cardiovascular and metabolic causes						
Heart disease (per 100,000; pre-treatment mean 156.6)	1.009	-4.693	1.703	38.00***	-0.0211	36.37***
	(3.980)	(4.743)	(3.725)	(2.145)	(0.0327)	(13.09)
% of pre-treatment mean	0.6%	-3.0%	1.1%	56.4%	-2.1%	23.1%
Cerebrovascular (per 100,000; pre-treatment mean 36.72)	-1.285	0.7241	-1.962	3.025***	0.0038	5.481
	(2.306)	(2.652)	(1.653)	(1.025)	(0.0700)	(9.859)
% of pre-treatment mean	-3.5%	2.0%	-5.3%	19.2%	0.4%	14.9%
Hypertension (per 100,000; pre-treatment mean 26.68)	-5.943***	1.136	-6.500***	-46.11***	-0.0707	
	(2.221)	(2.862)	(1.231)	(1.329)	(0.0859)	
% of pre-treatment mean	-22.3%	4.3%	-24.4%	-412.2%	-6.8%	
Diabetes (per 100,000; pre-treatment mean 51.55)	4.913**	5.095	4.444**	-2.263	0.1079**	-10.45
	(2.409)	(3.108)	(1.803)	(1.469)	(0.0479)	(9.981)
% of pre-treatment mean	9.5%	9.9%	8.6%	-10.3%	11.4%	-20.2%
Respiratory and infectious causes						
Pneumonia and influenza (per 100,000; pre-treatment mean 29.44)	0.9216	0.3048	1.645	6.417***	0.0068	
	(2.363)	(2.959)	(1.770)	(1.658)	(0.1143)	
% of pre-treatment mean	3.1%	1.0%	5.6%	47.8%	0.7%	
Cancer and chronic liver disease						
Cancer (per 100,000; pre-treatment mean 114.4)	-9.277**	-11.35**	-5.490**	3.878*	-0.0874*	20.16***
	(4.245)	(5.403)	(2.792)	(2.127)	(0.0507)	(5.241)
% of pre-treatment mean	-8.1%	-9.9%	-4.8%	7.6%	-8.4%	17.5%
Cirrhosis (per 100,000; pre-treatment mean 18.64)	-0.6440	0.1310	-0.7293	-4.479***	0.0043	107.2***
	(1.245)	(1.806)	(0.9615)	(0.8057)	(0.1131)	(14.10)
% of pre-treatment mean	-3.5%	0.7%	-3.9%	-55.6%	0.4%	573.9%
External causes						
Accidents (per 100,000; pre-treatment mean 36.02)	-1.533	1.804	-2.554	-4.770	-0.0903	6.410
	(2.433)	(2.832)	(2.800)	(4.360)	(0.0752)	(4.488)
% of pre-treatment mean	-4.3%	5.0%	-7.1%	-31.3%	-8.6%	17.7%
Unclassified						
Unclassified (per 100,000; pre-treatment mean 254.2)	38.79***	4.252	41.72***	179.5***	0.0754**	
	(6.722)	(8.877)	(7.432)	(11.48)	(0.0354)	
% of pre-treatment mean	15.3%	1.7%	16.4%	155.4%	7.8%	
Panel B. Prenatal-care-sensitive outcomes						
Infant mortality (per 1,000 births; pre-treatment mean 12.56)	0.0985	0.6985	-0.5230	6.531***	0.0170	
	(0.7941)	(1.077)	(0.6708)	(1.204)	(0.1013)	
% of pre-treatment mean	0.8%	5.6%	-4.2%	66.8%	1.7%	
Maternal mortality (per 100,000 births; pre-treatment mean 19.27)	-4.291	-19.70	8.345	0.7297***	1.362***	
	(22.43)	(25.65)	(7.469)	(0.0773)	(0.2559)	
% of pre-treatment mean	-22.3%	-102.2%	43.3%	578.5%	290.4%	
Treated counties	78	77	78	78	78	78
Never-treated counties						147

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. The never-treated column is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), the same covariates as the never-treated tables. Unclassified is all-cause deaths less the causes printed above it, so it closes the decomposition rather than naming a further cause. It also holds chronic pulmonary, septicemia, AIDS and homicide, which this table does not report separately. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10).

Table A7: Never-Treated Control Groups, Conditional

	(1) Non-MMC	(2) Non-HMO
All-cause mortality		
ATT	23.15	21.57
	(20.54)	(21.67)
Pre-treatment mean	731.8	731.8
ATT as % of pre-treatment mean	3.2%	2.9%
Treated counties	78	78
Never-treated counties	282	353

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The outcome is measured per 100,000. Puerto Rico against never-treated mainland counties, through 2000. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Never-treated mortality comes from CDC WONDER. All-cause deaths do not depend on cause coding, so the ICD-9 to ICD-10 revision does not break this series. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census). The island poverty rate is families below the Census threshold over all families; the mainland rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise.

Table A8: Never-Treated Control Groups: Mortality by Cause, Conditional

	(1) Non-MMC	(2) Non-HMO
Mortality		
All-cause mortality		
All causes (per 100,000; pre-treatment mean 731.8; 78 treated counties)	23.15	21.57
	(20.54)	(21.67)
% of pre-treatment mean	3.2%	2.9%
Never-treated counties	282	353
Cardiovascular and metabolic causes		
Heart disease (per 100,000; pre-treatment mean 157.7; 78 treated counties)	36.37***	38.68***
	(13.09)	(12.40)
% of pre-treatment mean	23.1%	24.5%
Never-treated counties	255	273
Cerebrovascular (per 100,000; pre-treatment mean 36.84; 78 treated counties)	5.481	-6.707
	(9.859)	(6.453)
% of pre-treatment mean	14.9%	-18.2%
Never-treated counties	148	146
Diabetes (per 100,000; pre-treatment mean 51.73; 78 treated counties)	-10.45	-10.11
	(9.981)	(6.793)
% of pre-treatment mean	-20.2%	-19.5%
Never-treated counties	65	56
Respiratory and infectious causes		
Pneumonia and influenza (per 100,000; pre-treatment mean 29.88; 78 treated counties)		-4.159
		(4.960)
% of pre-treatment mean		-13.9%
Never-treated counties		51
Cancer and chronic liver disease		
Cancer (per 100,000; pre-treatment mean 115.3; 78 treated counties)	20.16***	25.75***
	(5.241)	(5.650)
% of pre-treatment mean	17.5%	22.3%
Never-treated counties	188	194
Cirrhosis (per 100,000; pre-treatment mean 18.68; 78 treated counties)	107.2***	106.5***
	(14.10)	(14.49)
% of pre-treatment mean	573.9%	569.9%
Never-treated counties	255	273
External causes		
Accidents (per 100,000; pre-treatment mean 36.27; 78 treated counties)	6.410	0.9650
	(4.488)	(7.151)
% of pre-treatment mean	17.7%	2.7%
Never-treated counties	147	139

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico against never-treated mainland counties, through 2000. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Mainland deaths come from CDC WONDER in ICD-9 through 1998 and ICD-10 from 1999, a revision Puerto Rico crosses under a different coding system, so the cause-specific rows carry a source break that year effects do not remove. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census).

Table A9: Never-Treated Control Groups (until 2003), Conditional

	(1) Non-MMC	(2) Non-HMO
All-cause mortality		
ATT	16.22	14.93
	(19.39)	(27.23)
Pre-treatment mean	731.8	731.8
ATT as % of pre-treatment mean	2.2%	2.0%
Treated counties	78	78
Never-treated counties	282	353

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The outcome is measured per 100,000. Puerto Rico against never-treated mainland counties, through 2003. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Never-treated mortality comes from CDC WONDER. All-cause deaths do not depend on cause coding, so the ICD-9 to ICD-10 revision does not break this series. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census). The island poverty rate is families below the Census threshold over all families; the mainland rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise.

Table A10: Never-Treated Control Groups: Mortality by Cause (until 2003), Conditional

	(1) Non-MMC	(2) Non-HMO
Mortality		
All-cause mortality		
All causes (per 100,000; pre-treatment mean 731.8; 78 treated counties)	16.22	14.93
	(19.39)	(27.23)
% of pre-treatment mean	2.2%	2.0%
Never-treated counties	282	353
Cardiovascular and metabolic causes		
Heart disease (per 100,000; pre-treatment mean 157.7; 78 treated counties)	69.55***	59.73***
	(13.88)	(14.71)
% of pre-treatment mean	44.1%	37.9%
Never-treated counties	255	273
Cerebrovascular (per 100,000; pre-treatment mean 36.84; 78 treated counties)	-0.4586	-9.508
	(11.03)	(7.093)
% of pre-treatment mean	-1.2%	-25.8%
Never-treated counties	148	146
Diabetes (per 100,000; pre-treatment mean 51.73; 78 treated counties)	-10.00	-11.18
	(8.037)	(7.140)
% of pre-treatment mean	-19.3%	-21.6%
Never-treated counties	66	56
Respiratory and infectious causes		
Pneumonia and influenza (per 100,000; pre-treatment mean 29.88; 78 treated counties)		-7.954*
		(4.797)
% of pre-treatment mean		-26.6%
Never-treated counties		53
Cancer and chronic liver disease		
Cancer (per 100,000; pre-treatment mean 115.3; 78 treated counties)	31.46***	37.02***
	(5.493)	(6.269)
% of pre-treatment mean	27.3%	32.1%
Never-treated counties	188	194
Cirrhosis (per 100,000; pre-treatment mean 18.68; 78 treated counties)	183.8***	164.9***
	(16.31)	(19.09)
% of pre-treatment mean	983.5%	882.5%
Never-treated counties	255	273
External causes		
Accidents (per 100,000; pre-treatment mean 36.27; 78 treated counties)	11.70***	1.833
	(4.138)	(9.110)
% of pre-treatment mean	32.3%	5.1%
Never-treated counties	149	140

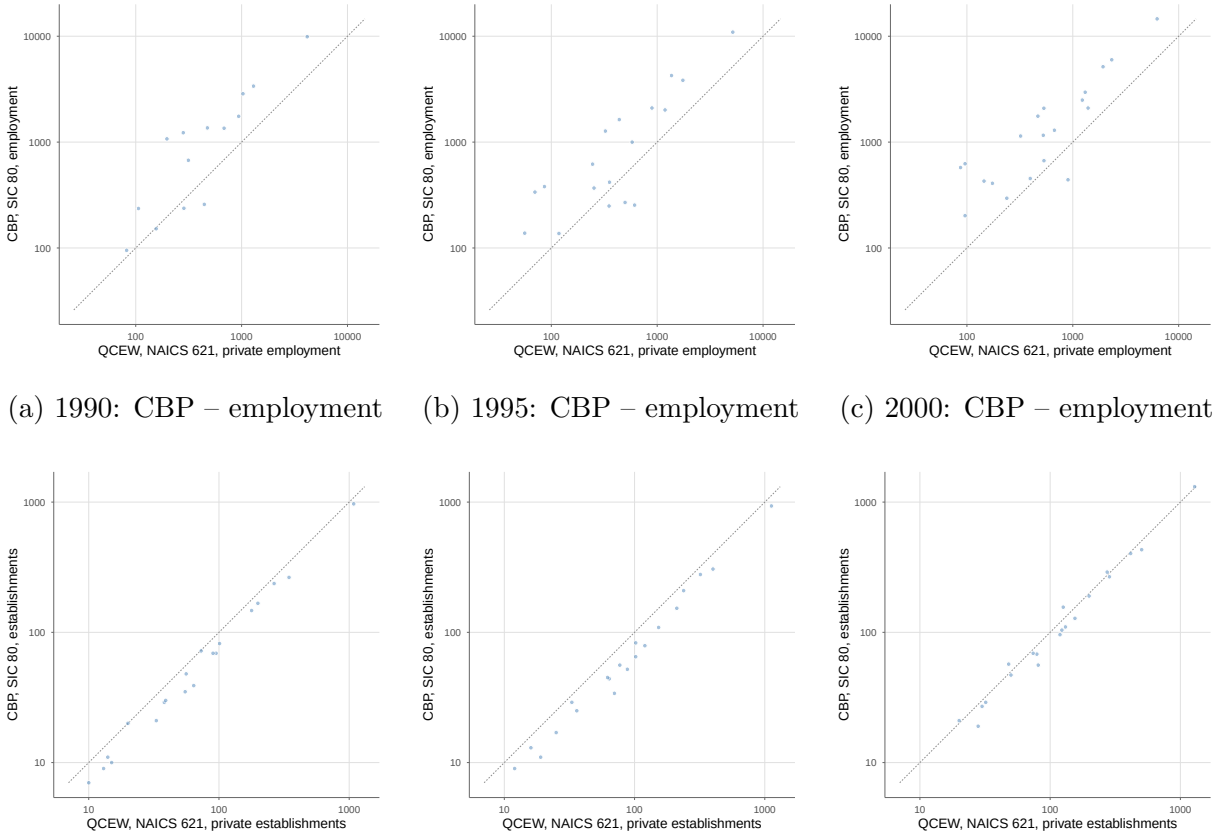
Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Puerto Rico against never-treated mainland counties, through 2003. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Mainland deaths come from CDC WONDER in ICD-9 through 1998 and ICD-10 from 1999, a revision Puerto Rico crosses under a different coding system, so the cause-specific rows carry a source break that year effects do not remove. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census).

Table A11: Health professionals per 1,000 residents and island totals by year

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Licensed health professionals (Puerto Rico Planning Board)											
Island totals, headcount											
Physicians, total			8,443			6,764			7,144		
Dentists, total			1,329			1,075			1,028		
Registered nurses, total			16,848			12,541			12,145		
Practical nurses, total			8,881			7,128			6,536		
Medical technologists, total			2,751			2,179			2,294		
Health professionals											
Physicians per 1,000 residents			1.061			0.980			1.047		
	(–)	(–)	(1.205)	(–)	(–)	(0.929)	(–)	(–)	(0.949)	(–)	(–)
Dentists per 1,000 residents			0.198			0.183			0.180		
	(–)	(–)	(0.164)	(–)	(–)	(0.121)	(–)	(–)	(0.119)	(–)	(–)
Registered nurses per 1,000 residents			2.240			1.981			1.897		
	(–)	(–)	(2.674)	(–)	(–)	(2.053)	(–)	(–)	(1.933)	(–)	(–)
Practical nurses per 1,000 residents			1.368			1.281			1.153		
	(–)	(–)	(1.340)	(–)	(–)	(1.129)	(–)	(–)	(0.984)	(–)	(–)
Medical technologists per 1,000 residents			0.375			0.346			0.373		
	(–)	(–)	(0.398)	(–)	(–)	(0.312)	(–)	(–)	(0.321)	(–)	(–)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Workforce counts are licensed professionals, published every third year, so blank columns are uncovered years rather than zeros.

Figures

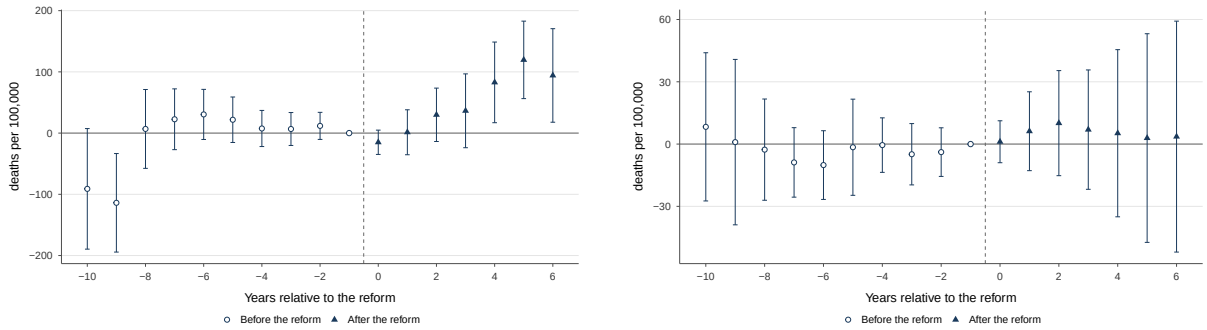


(a) 1990: CBP – employment (b) 1995: CBP – employment (c) 2000: CBP – employment

(d) 1990: CBP – establishments (e) 1995: CBP – establishments (f) 2000: CBP – establishments

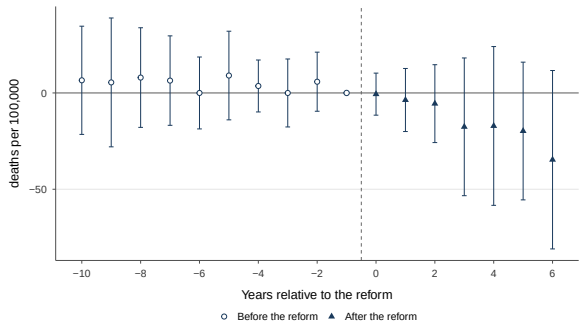
Figure A1: QCEW compared to CBP

Each point is one county in the year the panel names, over the counties both sources publish a positive figure for. QCEW is NAICS 621, ambulatory health care services, restricted to private ownership; CBP is SIC 80, all health services, so it also carries hospitals and nursing and residential care. CBP counts a single reference period and QCEW an annual average. Counties drawn: employment 14 to 20 per year, establishments 20.

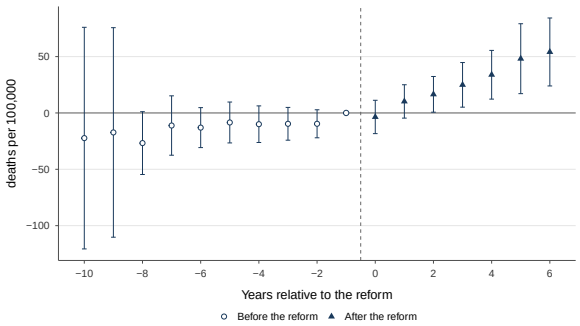


(a) Heart disease

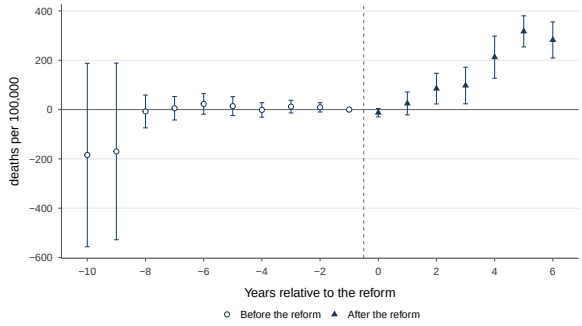
(b) Cerebrovascular disease



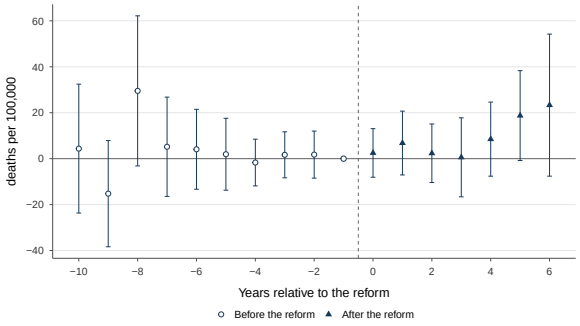
(c) Diabetes



(d) Cancer



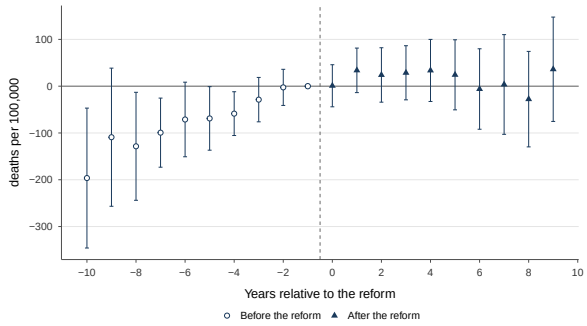
(e) Chronic liver disease



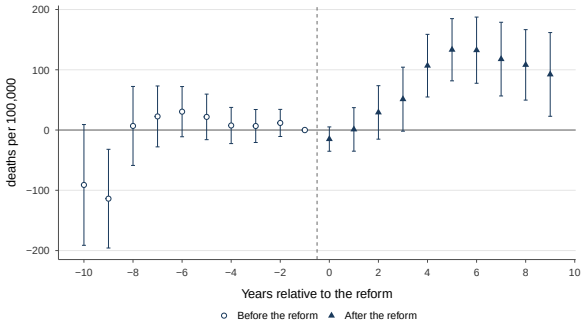
(f) Accidents

Figure A2: Never-treated comparison, mortality by cause

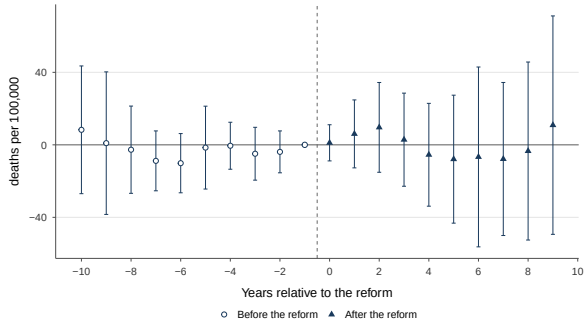
Callaway and Sant'Anna (2021) estimates by year relative to adoption, comparing Puerto Rico counties with never-treated mainland counties in states that had not adopted Medicaid managed care by the end of the window, conditional on population density (1990 to 1993 mean) and the 1990 Census poverty rate, through 2000. Mainland deaths come from CDC WONDER, which codes them in ICD-9 through 1998 and in ICD-10 from 1999. The year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



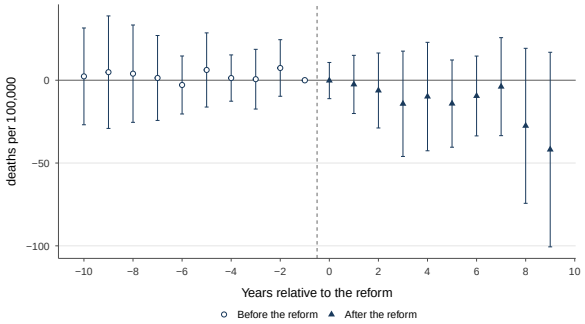
(a) All causes



(b) Heart disease

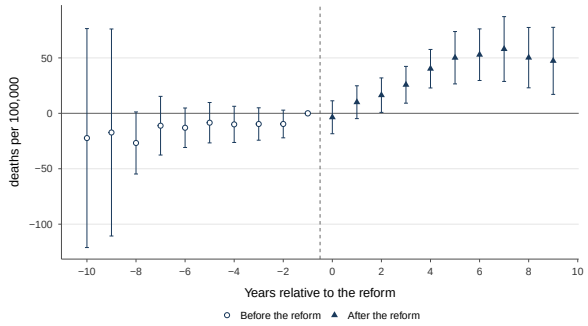


(c) Cerebrovascular disease

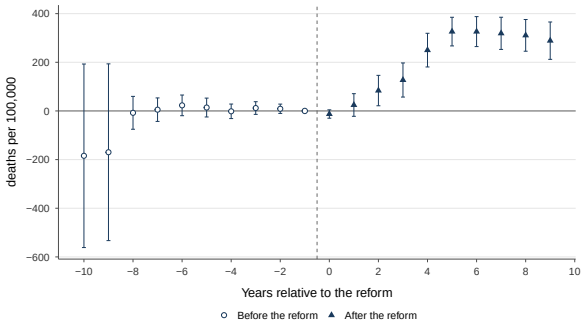


(d) Diabetes

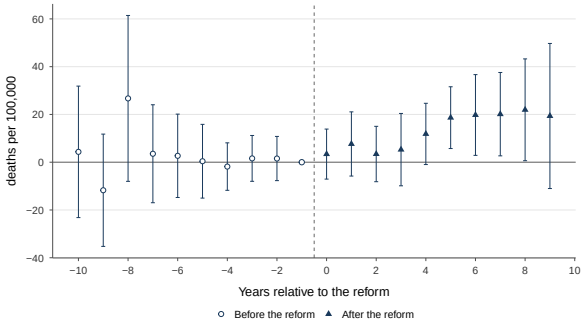
Figure A3: Never-treated comparison through 2003



(e) Cancer



(f) Chronic liver disease



(g) Accidents

Figure A3: Never-treated comparison through 2003 (continued)

Callaway and Sant'Anna (2021) estimates by year relative to adoption, comparing Puerto Rico counties with never-treated mainland counties in states that had not adopted Medicaid managed care by the end of the window, conditional on population density (1990 to 1993 mean) and the 1990 Census poverty rate, through 2003. Mainland deaths come from CDC WONDER, which codes them in ICD-9 through 1998 and in ICD-10 from 1999. All-cause deaths do not depend on cause coding. The year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

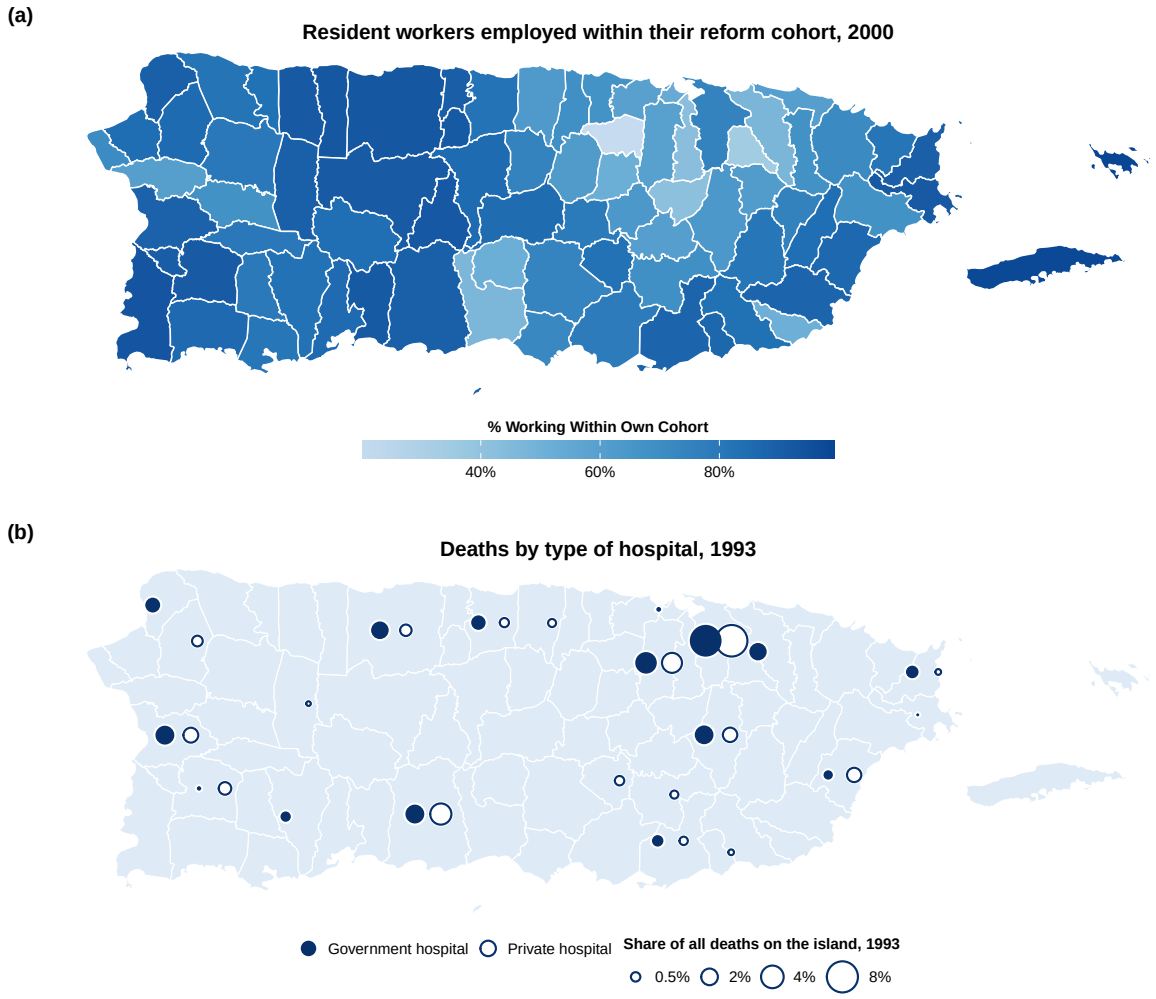


Figure A4: Commuting within reform cohorts, 2000, and deaths by type of hospital, 1993

The commuting map shades each county by the share of its resident workers who worked in a county of the same reform cohort, their own county included. Across the island in 2000, 52.5 percent of resident workers worked in their own county, 18.1 percent in another county of the same cohort and 29.3 percent in a county of a different cohort. The hospital map places a circle on each county whose hospitals recorded deaths in 1993, filled for government hospitals and open for private ones, sized by the share of all 28,494 deaths on the island that occurred there; together, government hospitals recorded 30.5 percent of the island's deaths and private hospitals 24.0 percent. Every cohort had at least one county with each kind of hospital: of the 18 counties of the 1994 cohort, 4 had a government hospital and 3 a private one; of the 27 of the 1995 cohort, 2 and 6; of the 7 of the 1996 cohort, 1 and 1; of the 13 of the 1997 cohort, 4 and 3; of the 12 of the 1998 cohort, 4 and 3; and San Juan, the 2000 cohort, had both. Sources: U.S. Census Bureau, Census 2000 county-to-county worker flows, summed from minor civil divisions; Puerto Rico Department of Health, Annual Vital Statistics Report 1993, Table 10.

B Reform financing and the spending series

This appendix defines the spending series in Figure 1, distinguishing the financing of the government health plan from delivery costs, and lists the plan's covered services. The island-level trends describe the reform period but do not identify its causal effects.

The reform redirected public financing toward premiums paid through the Health Insurance Administration to private insurers. Government funding continued for retained facilities and other public functions. Fiscal year 2002–2003 financing of the government health plan totaled about $2.24 billion in constant 2025 dollars: roughly 73 percent from the Commonwealth General Fund, 15 percent from federal Medicaid, 11 percent from counties, and 2 percent from CHIP (Pan American Health Organization, 2007). Operating-budget volumes yield separate estimates of $2.27 billion in fiscal 2002 and $2.18 billion in fiscal 2003 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 2004). Federal Medicaid funding was subject to a territorial cap and matching rules that differed from the income-based matching formula for states (Medicaid and CHIP Payment and Access Commission, 2020; Pillai et al., 2024).

Between fiscal years 1996 and 1999, the Puerto Rico General Fund support for health delivery and regulation, excluding the Health Insurance Administration, fell from about $985 million to $572 million. Own-source revenue rose from about $382 million to $483 million, while federal funding remained near $650 million. Budgeted positions fell from 26,905 to 18,062, or 33 percent (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998).

The public-delivery series covers spending by government-operated agencies, the health services assistance fund, the Medical Center, and the Cardiovascular Center. It comes from financial statements and falls below the operating-budget totals reported elsewhere. The private-delivery series records government premiums paid to managed-care organizations, excluding household payments and privately financed insurance. Their sum covers the gov-

ernment spending represented by these two series. Household medical spending is instead a national-accounts share of personal consumption. That share rose from about 5.6 percent in 1971 to 12.4 percent in 1993 and 17.4 percent by fiscal year 2000 (Junta de Planificación de Puerto Rico, 2024).

The government health plan's benefit package covered inpatient and outpatient hospital care, laboratory and x-ray services, vision care, immunization, family planning, transportation, physician and ambulatory services, pharmacy, dental care, mental-health care, and substance-abuse treatment, without pre-existing-condition exclusions (Centers for Medicare and Medicaid Services, 2000).

C Framework derivation

This appendix sets out and solves the model summarized in Section 3 and proves Proposition 1.

Following Hart et al. (1997), a government representing society contracts with a delivery manager over a basic service bundle at a fixed payment, which is a salary under public control and a fixed amount per enrollee under private control. I treat the insurer and its providers as one decision maker, which assumes that the insurer passes the incentive of its capitated payment through to the providers it contracts with. Public or private control determines who can implement changes outside the initial contract. Eligibility, cost sharing, and covered benefits are held fixed.

For service s, cost-reducing effort $e_s \geq 0$ generates production savings $c_s(e_s)$ and patient harm $b_s(e_s)$. Quality-improving effort $i_s \geq 0$ generates value to patients $v_s(i_s)$ at direct production cost $k_s(i_s)$, so its net benefit is

$$\beta_s(i_s) = v_s(i_s) - k_s(i_s). \tag{C.1}$$

All components are measured in common value units and vanish at zero. The functions c_s and β_s are increasing and strictly concave, v_s is increasing, and b_s has positive, nondecreasing marginal harm. Assume $c_s' - b_s' \geq 0$ over the relevant effort range, so cost innovations remain worth implementing. Innovations are observable to the parties but noncontractible. The parties are risk neutral and split renegotiation surplus equally. With managerial effort costs normalized to $e_s + i_s$, the first best maximizes incremental social surplus:

$$\max_{\{e_s, i_s\}_{s \in \mathcal{S}}} \sum_{s \in \mathcal{S}} \left[c_s(e_s) - b_s(e_s) + \beta_s(i_s) - e_s - i_s \right]. \tag{C.2}$$

A private manager retains cost savings without bearing the associated patient harm and receives half the net benefit of quality innovation through renegotiation. Under public control, both innovations require the government's approval, and the government values a cost innovation net of the patient harm. If the share $\lambda \in (0, 1]$ of innovation value requires the incumbent manager's participation, replacement allows the government to obtain the remainder, so bargaining gives the manager $\lambda/2$ of net innovation value. These payoffs give the private and public managers' problems (1) and (2) in Section 3.

Assuming differentiability and interior optima, the first-order conditions for the first best ($*$), private control (P), and public control (G) are

$$\begin{aligned} c_s'(e_s^*) - b_s'(e_s^*) &= 1, & \beta_s'(i_s^*) &= 1, \\ c_s'(e_s^P) &= 1, & \tfrac{1}{2}\beta_s'(i_s^P) &= 1, \\ \tfrac{\lambda}{2}\left[c_s'(e_s^G) - b_s'(e_s^G) \right] &= 1, & \tfrac{\lambda}{2}\beta_s'(i_s^G) &= 1. \end{aligned} \tag{C.3}$$

The public cost condition requires $c_s' - b_s' = 2/\lambda$, which exceeds its first-best value of 1, and $c_s' - b_s'$ declines with effort, so $e_s^G < e_s^*$. The private condition $c_s' = 1$ omits the marginal harm; at e_s^* the first-best condition gives $c_s' = 1 + b_s' > 1$, and c_s' declines with effort, so $e_s^P > e_s^*$. The quality conditions set the marginal net benefit of quality equal to 1 at the first

best, 2 under private control, and $2/\lambda \geq 2$ under public control, and β_s' declines with effort, so $i_s^G \leq i_s^P < i_s^*$. Together these conditions order the efforts:

$$e_s^G < e_s^* < e_s^P, \qquad i_s^G \leq i_s^P < i_s^*, \tag{C.4}$$

with $i_s^G = i_s^P$ when $\lambda = 1$. As $\lambda \to 0$, both public efforts go to zero. Private control pushes cost-reducing effort past the first best, because the private manager ignores the harm, and raises quality-improving effort above the public level whenever $\lambda < 1$. Greater cost-saving effort need not reduce total production costs, since quality improvement may require additional resources.

To connect effort to mortality, let $a_s \geq 0$ measure how much service s matters for survival, and write mortality under regime $r \in \{G, P\}$ as

$$m^r = \bar{m} - \sum_s a_s \left[v_s(i_s^r) - b_s(e_s^r)\right] + u, \tag{C.5}$$

where u collects the determinants of mortality outside healthcare, which do not depend on the control regime. In the data u varies across counties and years, and the identifying assumptions of Section 5 require its average change to be the same in counties that entered as in counties that had not yet entered.

Proposition 1 *Let $a_s > 0$ for some service, and let*

$$\Delta m \equiv m^P - m^G = \sum_s a_s \Big\{ \big[b_s(e_s^P) - b_s(e_s^G)\big] - \big[v_s(i_s^P) - v_s(i_s^G)\big] \Big\}. \tag{C.6}$$

(a) If $\lambda = 1$, mortality is higher under private control: $\Delta m > 0$. (b) For $\lambda < 1$, $\Delta m > 0$ if and only if $\sum_s a_s b_s'(\tilde{e}_s)(e_s^P - e_s^G) > \sum_s a_s v_s'(\tilde{\imath}_s)(i_s^P - i_s^G)$, where $\tilde{e}_s$ lies between e_s^G and e_s^P and $\tilde{\imath}_s$ between i_s^G and i_s^P. Both effort gaps narrow as λ rises, and the quality gap closes at $\lambda = 1$. Because Δm is continuous in λ and positive at $\lambda = 1$, it is positive on an interval

$(\underline{\lambda}, 1]$.

Mortality rises under private control when, weighted by each service's importance for survival, the harm from additional cost-cutting outweighs the value to patients of additional quality innovation. Privatization brings a quality gain only to the extent that public managers captured less of an innovation's value than private ones ($\lambda < 1$), so higher mortality is the prediction when public managers were hard to replace and remains possible otherwise. The parameter λ, the survival weights a_s, and the value functions v_s are not observed, so the condition in part (b) cannot be tested.

Proof of Proposition 1. Substituting (C.5) for each regime and differencing gives (C.6). (a) At $\lambda = 1$ the two quality conditions in (C.3) coincide, so $i_s^G = i_s^P$ and the quality term is zero. The cost conditions give $c_s'(e_s^P) = 1 < 2 + b_s'(e_s^G) = c_s'(e_s^G)$, so $e_s^G < e_s^P$ because c_s' is decreasing, and $b_s' > 0$ gives $b_s(e_s^P) > b_s(e_s^G)$ for every s. With $a_s \geq 0$ and $a_s > 0$ for some s, $\Delta m > 0$. (b) By the mean value theorem, $b_s(e_s^P) - b_s(e_s^G) = b_s'(\tilde{e}_s)(e_s^P - e_s^G)$ and $v_s(i_s^P) - v_s(i_s^G) = v_s'(\tilde{\imath}_s)(i_s^P - i_s^G)$ for intermediate points $\tilde{e}_s$ and $\tilde{\imath}_s$, which gives the condition. Differentiating the public conditions gives $de_s^G/d\lambda = -2/[\lambda^2(c_s'' - b_s'')] > 0$ and $di_s^G/d\lambda = -2/(\lambda^2\beta_s'') > 0$, while the private efforts do not depend on λ, so both gaps narrow as λ rises and the quality gap closes at $\lambda = 1$. The public efforts are continuous in λ, so Δm is continuous, and $\Delta m > 0$ at $\lambda = 1$ implies $\Delta m > 0$ on some interval $(\underline{\lambda}, 1]$. ■

To connect effort to employment and capacity, let the quantity of input j, such as staff or facilities, used in service s be

$$x_{js} = \bar{x}_{js} - \ell_{js}(e_s) + \kappa_{js}(i_s), \qquad \ell_{js}' \geq 0. \tag{C.7}$$

Here $\ell_{js}(e_s)$ is the quantity that cost-reducing effort saves and $\kappa_{js}(i_s)$ the change that quality improvement brings, both zero without effort, so $\bar{x}_{js}$ is the quantity used without either innovation. The sign of κ_{js}' is left free, so quality improvement can require the input ($\kappa_{js}' > 0$) or save it ($\kappa_{js}' < 0$). Inputs decline under private control when the additional savings from

containment exceed the additional requirements of quality improvement, and they never rise when quality improvement also saves the input. A shortfall in employment or establishments is therefore consistent with either kind of innovation, whether it comes through slower hiring, attrition, establishment exit, or layoffs, and it measures neither effort nor efficiency.

Early prenatal care measures timely utilization, while birth outcomes and mortality reflect downstream health consequences. To illustrate how sustained changes in care can accumulate, let health investment $I(q_t)$ depend on effective medical care q_t, with $I' > 0$ and depreciation $\delta \in (0, 1)$ (Grossman, 1972):

$$H_{t+1} = (1 - \delta)H_t + I(q_t), \qquad \Delta H_t = \Delta I \, \frac{1 - (1 - \delta)^t}{\delta}. \tag{C.8}$$

The second expression compares paths with identical initial health and a permanent investment difference ΔI beginning at $t = 0$. Reduced investment progressively lowers health relative to the counterfactual. Translating this decline into mortality requires an additional relationship between health and survival; the equation alone establishes neither a mortality lag nor a ranking of diseases. Each cause of death can depend on several services, so cause-specific mortality does not identify the quality of a corresponding service.

D Data construction

This appendix records panel construction in the source order of Section 4. These checks assess data consistency and comparability, not the identifying assumptions.

The vital-statistics complement is transcribed from annual report scans. Each table is checked against printed area subtotals, island totals, and internal decompositions: government, private, and other institutional births against the place-of-birth total; and attended, unattended, and unspecified counts against institutional births. Uncertain cells are inspected in enlarged scans; missing cells are not inferred. Two exceptions remain. Because the 1990

natality table is absent from the image set, low-birth-weight and very-low-birth-weight counts use an earlier extraction, checked only to ensure each does not exceed births. In four counties, place-of-birth rows differ from Planning Board births by one or two; these discrepancies are retained. Replication documentation records the protocol, annual checks, and patches. Agreement between overlapping publications establishes consistency, not independent registration coverage.

Merging the vital-statistics complement. Source flags trace each observation in the merged Planning Board and digitized reports to its publication. Only counts are merged; rates are calculated with the panel's own denominators. Digitized records supplement early coverage of the four causes incompletely reported by the Planning Board, including observations from 1990 to 1995.

Industry definitions. The three healthcare employment series in Appendix Table A2 cover different industries and workers. The QCEW series is NAICS 621, Ambulatory Health Care Services, which leaves out hospitals (NAICS 622) and nursing and residential care (NAICS 623), and it counts jobs covered by unemployment insurance at their workplace (U.S. Bureau of Labor Statistics, 2025). County Business Patterns reports SIC 80, the health services division as published, which covers offices of physicians, dentists and other practitioners (801–804), nursing and personal care (805), hospitals (806), laboratories (807), and allied services (809). Its employment refers to the pay period including March 12, and its components are published separately in too few counties to be netted out of the division, so its series is broader than the ambulatory ones (U.S. Census Bureau, 2000). The Census series counts employed civilians aged 16 and over, the self-employed and workers outside unemployment-insurance coverage included, in the ambulatory categories of the health services group of the 1990 Census industrial classification: offices and clinics of physicians (812), dentists (820), chiropractors (821), optometrists (822), and health practitioners n.e.c. (830),

and health services n.e.c. (840). It leaves out the group's hospitals (831) and nursing and personal care facilities (832), and its counts are weighted estimates from the Puerto Rico five-percent microdata samples of 1990 and 2000 (Ruggles et al., 2025).

E Institutional detail

This appendix reports the institutional detail that Section 2 summarizes.

The public delivery system before the reform. CDTs offered primary and emergency care, laboratory testing, radiology, and pharmacy services; by 1993, many also offered internal medicine, obstetrics and gynecology, and pediatrics. They housed vaccination, communicable-disease, and environmental-health programs, alongside federally funded maternal and child health and agricultural migrant programs. Family Health Centers added dental care, social work, nutrition, and epidemiological nursing, with ambulatory teams comprising a physician, public-health nurse, and health assistant. Sub-regional and regional hospitals supplied inpatient and specialized outpatient care and housed many accredited residency programs (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Patients entered through the public health center of their county and were referred from there to an area or regional hospital and, when necessary, to one of three medical centers (Pagán-Berlucchi and Muse, 1983).

Medicaid before the reform. Medicaid reimbursed the Department of Health an all-inclusive rate per inpatient day or outpatient visit, with no fee-for-service payment to physicians, and its recipients carried a Medicaid card. Public funds paid for Medicaid's categorically and medically needy recipients and for a third group of indigent residents who were ineligible for Medicaid and lacked health insurance, whose care a 1969 Commonwealth law guaranteed with Commonwealth funds alone. In 1979, about 1.3 million of the island's roughly 3.2 million residents were eligible for and enrolled in Medicaid. That year the island had 30,462 practicing health professionals, 5,368 of them physicians, and most physicians practiced privately. Public physicians earned about $15,000 to $30,000 a year in early-1980s dollars, against an estimated $45,000 to $50,000 average in private practice (Pagán-Berlucchi and Muse, 1983).

Who relied on the public system in 1990. In 1990, about 1.2 million residents had private insurance, 0.3 million had Medicare Parts A and B, and 0.2 million were public employees whose insurance premium the government partly paid. The other 1.8 million relied on public care, about 0.9 million of them through federal Medicaid, 0.3 million through the Commonwealth's own program and 0.6 million enrolled in neither. The Department of Health had about 1,000 physicians for these 1.8 million residents, while about 7,000 physicians served the other 1.7 million privately (Pan American Health Organization, 2007).

Where government and private hospitals were. In 1993, deaths occurred in government hospitals in 16 of the 78 counties and in private hospitals in 17, 22 counties in all, and the government and private hospitals of San Juan alone recorded 8.4 and 8.3 percent of all deaths on the island (Puerto Rico Department of Health, 2025); Appendix Figure A4 maps these shares by county. In 1992, government hospitals accounted for 39,418 of the island's 64,137 hospital births, or 61.5 percent (Puerto Rico Department of Health, 2025).

Facility transfers and provider payment. The transfers began in 1994. Act 190 of September 5, 1996, did not affect the contracts signed before it and governed every later lease, sublease, sale, cession or other contract transferring the Department of Health's facilities to private interests. When the Secretary of Health chose the areas the government health plan would enter, the Department requested proposals for its facilities there. A provider taking over a facility had to commit to accept the coverage of the insurer the Health Insurance Administration contracted, and the Department's regular employees who did not join the new provider kept their rights under the Executive Reorganization Act of 1993 (Commonwealth of Puerto Rico, 1996). The CDTs were sold to private medical groups or nonprofit corporations, which became independent practice associations. By 2001, after many of the emergency rooms the CDTs had run closed, 58 of the 78 counties had one open around the clock (Pan American Health Organization, 2007). Plans placed financial risk on primary-care physicians

for the care other providers gave their enrollees, which Act 72 had not provided for, and the commission that evaluated the system in 2005 found that this risk faced primary-care physicians with the possibility of denying indicated services or delaying clinically necessary tests (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005; Pan American Health Organization, 2007).

The retained Medical Center. The Administration of Medical Services, which runs centralized clinical and support services for the Medical Center's hospitals, filled 2,068 positions across its three programs in December 1992 and 1,998 in September 1999 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).

Plans and the order of entry. Each county's residents enrolled in the one plan serving their county, so enrollees had no choice of plan, and every plan covered the benefits the government defined (World Health Organization and Pan American Health Organization, 1998; Brugueras Fabre, 2002; Pan American Health Organization, 2007). The sources give the schedule of entry but not why the counties entered in that order. The 1998 cohort and San Juan had the largest populations and densities at baseline, while the 1994 cohort had the highest government share of births and more hospitals per resident than any cohort entering between 1995 and 1998 (Appendix Table A1). A county's cohort year is the first year in which the Health Insurance Administration counted enrollees in its region, so a county the reform reached late in a year is treated for that whole year.

Eligibility, take-up and coverage after the reform. The commission that evaluated the health system in 2005 describes the 200 percent income limit as a rise from 100 percent of the poverty line (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Among eligible residents, take-up was 86 percent in 1995 and 1996, 93–98 percent from 1997 through 2005, and 90 percent in 2006 and 2007 (Administración de Seguros de Salud de Puerto Rico, 2026). Mental-health and pharmacy administration were

subsequently contracted separately (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

Coverage before and after the reform. The table below compares the coverage of residents in 1990 and 2003. About half of residents were in a public program or had no coverage in both years, 51 percent in 1990 and 47 percent in 2003, and about a third held private insurance. Enrollment in the government health plan peaked at 1,795,067 in December 2000 and was 1,521,848 in December 2003, as the number of eligible residents fell from 1,866,965 to 1,618,540 (Administración de Seguros de Salud de Puerto Rico, 2026). The 2003 count is about 39 percent of the 3.879 million residents that Pan American Health Organization (2007) reports for that year, close to the commission's 40 percent. Pan American Health Organization (2007) puts the 2003 share without any insurance at about 8 percent, and a separate household survey estimated 353,058 uninsured residents, or 9.10 percent of the non-institutionalized population (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

Coverage of residents, 1990 and 2003

	1990		2003
Coverage	Millions	Percent	Percent
Private insurance	1.2	34	32
Medicare	0.3	9	14
Public employees' plan	0.2	6	12
Medicaid or the Commonwealth program (1990); the government health plan (2003)	1.2	34	40
None of these	0.6	17	7
Total	3.5	100	105

Notes: The 1990 counts are the Planning Board's, from its 1996 economic report as Pan American Health Organization (2007) reports them, rounded to 0.1 million; the percentages are of their 3.5 million total. The 2003 percentages are the commission's (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). They sum to 105 because about 5 percent of residents, eligible for both, are counted in the government health plan and in Medicare, and public employees enrolled in it are counted there (Pan American Health Organization, 2007). In 1990, the residents with none of these could use the public facilities. The sources do not say whether the public employees' figures include dependents.

Training and the licensed workforce. Accredited residency programs in primary-care specialties fell from 32 to 23 over the decade, partly because ambulatory training sites were lost when hospitals and CDTs were transferred to private providers (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Between the Planning Board counts of 1992 and 1995, licensed physicians fell from 8,443 to 6,764 and registered nurses from 16,848 to 12,541 (Appendix Table A11). I treat the three Planning Board waves as counted on one definition.